

General Topology

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standard Euclidean metric and generalized metrics, open and closed sets, the basis of a topology, subbasis, and other properties		It's Tikhonov babyyyy!	
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(12/20) Section 1-5,7 are done (for now).

(12/25) First 8 sections are done. 9 is done except for Arzela.

(12/26) Moved cell complex back to the main content. It's almost ready!

(1/10) Done.

(6/1) Added the "intro to AT" section. The appendix is finished (and fixed).

General Notice: The appendix is based on *Topology* by Bradley, Tai-Danae, et al.

These notes are adapted from professor I. Mineyev's lecture, FA23.

If you spot any errors in these notes, please reach out to me via email at b192.168.1.91.03@gmail.com, or send me a direct message on Discord at [3fsg](#).

*Mathematics is the art of giving
the same name to different things.*

——Henri Poincaré

Topological Spaces and Basis

Defined the standard Euclidean metric and generalized the notion of "metric" to other spaces. Discussed concepts such as open and closed sets in topologies, the basis of a topology, subbasis, and the properties that define a topological space. Lastly, we compared topologies.

Recall the standard Euclidean metric on $\mathbb{R}^2$. The Euclidean distance $d_{\text{Euc}} : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow [0, \infty)$ is a function defined as $d_{\text{Euc}}(x, y) := ((x_1 - y_1)^2 + (x_2 - y_2)^2)^{1/2}$. We can generalize this function to $\mathbb{R}^n$, where $n < \infty$, by $d_{\text{Euc}} : \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, \infty)$ such that $d_{\text{Euc}}(x, y) := (\sum_{i=1}^n (x_i - y_i)^2)^{1/2}$.

Vocabulary 1.1 (metric space). A metric is a distance measuring function $d : X \times X \rightarrow [0, \infty)$ satisfying

1. $d(x, y) = 0$ if and only if $x = y$ for all $x, y \in X$; (nondegeneracy)
2. $d(x, y) = d(y, x)$ for all $x, y \in X$; (symmetry)
3. $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$. (triangular inequality)

A pair (X, d) of a set X and a corresponding metric d is called a metric space.

Vocabulary 1.2 (open set of a metric space). Let X be a metric space with a metric d . An open ball at center x_0 with radius r is the set $B(x_0, r) := \{y \in X \mid d(y, x_0) < r\}$. An open set in X is a union of open balls in X .

Example. Open balls in $\mathbb{R}$ are open intervals. Open sets in $\mathbb{R}$ are disjoint unions of open intervals.

Proposition 1. Let X be a metric space with respect to the metric d , the open ball $B(x_0, r)$ is open.

Proof. Take any point $y \in B(x_0, r)$. Take $\rho = (r - d(x_0, y))/2$; then $B(y, \rho) \subseteq B(x_0, r)$. □

Example. Consider the boundary of the unit open ball centered at the origin of dimension $n+1$. The n -dimensional sphere, S^n , is defined as $S^n := \{y \in \mathbb{R}^{n+1} \mid d(y, 0) = 1\}$.

Remark. Let $\mathcal{F}$ be a collection of subsets of X . Denote

$$\bigcup \mathcal{F} := \bigcup_{U \in \mathcal{F}} U = \{y \in X \mid \exists U \in \mathcal{F} \text{ such that } y \in U\} \quad \text{and} \quad \bigcap \mathcal{F} := \bigcap_{U \in \mathcal{F}} U = \{y \in X \mid \text{for all } U \in \mathcal{F}, y \in U\}$$

Let $\mathcal{B}$ denotes the collections of all open balls in $\mathbb{R}^n$, that is, $\mathcal{B} = \{B(x, r) \mid x \in \mathbb{R}^n, r \in [0, \infty)\}$. Let $\mathcal{B}'$ be any subcollection of $\mathcal{B}$. Let $\mathcal{T} = \bigcup \mathcal{B}' = \{\bigcup_{B \in \mathcal{B}'} B \mid \mathcal{B}' \subseteq \mathcal{B}\}$. Given $U_1, U_2 \in \mathcal{T}$, where $U_1 = \bigcup_{U \in \mathcal{B}_1} U$ and $U_2 = \bigcup_{U \in \mathcal{B}_2} U$. We have $U_1 \cup U_2 \in \mathcal{T}$ since $U_1 \cup U_2 = (\bigcup_{U \in \mathcal{B}_1} U) \cup (\bigcup_{U \in \mathcal{B}_2} U) = \bigcup_{U \in \mathcal{B}_1 \cup \mathcal{B}_2} U$. It is true that $\bigcup i \in I U_i$ for an arbitrarily large index set. Also, $U_1 \cap U_2 \in \mathcal{T}$ since $U_1 \cap U_2 = (\bigcup_{U \in \mathcal{B}_1} U) \cap (\bigcup_{U' \in \mathcal{B}_2} U') = \bigcup_{U \in \mathcal{B}_1, U' \in \mathcal{B}_2} (U \cap U') \in \mathcal{T}$.

Vocabulary 1.3 (topological space). A topology on a set X is a collection $\mathcal{T}$ of subsets of X such that:

1. the set $\emptyset$ and X are in $\mathcal{T}$;
2. the union of the elements of any subcollection of $\mathcal{T}$ is in $\mathcal{T}$;
3. the intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

A topological space is a pair $(X, \mathcal{T})$, where $\mathcal{T}$ is a topology on X .

Vocabulary 1.4 (open set of a topological space). A subset U of X is called an open set if $U \in \mathcal{T}$.

Example. Let X be a set, then the collection of all subsets of X is trivially a topology on X , called the **discrete topology**. The collection $\{X, \emptyset\}$ is also a topology of X , called the **trivial topology**.

Vocabulary 1.5 (finer/coarser topology). Suppose that $\mathcal{T}$ and $\mathcal{T}'$ are two topologies on a given set X . If $\mathcal{T} \subseteq \mathcal{T}'$, we say that $\mathcal{T}'$ is finer than $\mathcal{T}$; if $\mathcal{T}'$ properly contains $\mathcal{T}$, we say that $\mathcal{T}'$ is strictly finer than $\mathcal{T}$. We say that $\mathcal{T}$ is *coarser* than $\mathcal{T}'$, or strictly coarser, in these two respective situations. We say $\mathcal{T}$ is comparable with $\mathcal{T}'$ if either $\mathcal{T} \subseteq \mathcal{T}'$ or $\mathcal{T}' \subseteq \mathcal{T}$.

Example. In $\mathbb{R}^n$, the union of open balls is called open sets. The collection of such open sets in $\mathbb{R}^n$ forms a topology.

Vocabulary 1.6 (neighborhood). Let X be a topological space. A neighborhood of $x \in X$ is an open set $U \subseteq X$ containing x .

Vocabulary 1.7 (Hausdorff). A topological space $(X, \mathcal{T})$ is Hausdorff if for all $x, y \in X$ and $x \neq y$, there exists some open sets U_1 and U_2 in $\mathcal{T}$ such that $x \in U_1$, $y \in U_2$, and $U_1 \cap U_2 = \emptyset$.

Vocabulary 1.8 (basis). A basis for a topology on a set X is a collection $\mathcal{B}$ of subsets of X . Basis elements satisfy:

1. for all $x \in X$, there exists a basis element B such that $x \in B$;
2. if $x \in B_1 \cap B_2$, where $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

The topology $\mathcal{T}$ generated by $\mathcal{B}$ consists of subsets $U \subseteq X$ if for all $x \in U$, there exists a basis element $B \in \mathcal{B}$ such that $x \in B$ and $B \subseteq U$.

Remark. Let $\mathcal{B}$ be a basis for some $\mathcal{T}$ on a set X , then $\emptyset, X \in \mathcal{B}$ by the definition. Consider $\{U_i\}_{i \in I}$ be a family of sets in $\mathcal{T}$, then for all $x \in \bigcup_{i \in I} U_i$, $x \in U_n$ for some n , then there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U \subseteq \bigcup_{i \in I} U_i$; for all $x \in \bigcap_{i=1}^n U_i$, there exists $x \in B_i \subseteq U_i$, where $B_i \in \mathcal{B}$, by induction, there exists $B \in \mathcal{B}$ such that $B \subseteq \bigcap_{i=1}^n B_i$, and B is the desired basis element. Hence $\mathcal{T}$ is a topology on X . Let $\mathcal{T}$ be the topology generated by the basis, denoted $\mathcal{B}$, and $\mathcal{T}'$ be the discrete topology on X . For all $U \in \mathcal{T}'$ and $x \in U$, then $x \in \{x\} \subseteq U$, hence $\mathcal{T}' \subseteq \mathcal{T}$. It is obvious that $\mathcal{T}'$ is the finest topology on X , then $\mathcal{T} \subseteq \mathcal{T}'$. Hence, $\mathcal{T} = \mathcal{T}'$.

Proposition 2. Let X be a set and $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on X . Then $\mathcal{T}$ equals the collection of all unions of elements of $\mathcal{B}$.

Proof. Let $\mathcal{T}'$ be the collection of all unions of elements of $\mathcal{B}$. Since $\mathcal{T}$ is generated by $\mathcal{B}$, $\mathcal{B} \subseteq \mathcal{T}$, then for all $U \in \mathcal{T}'$, $U \in \mathcal{T}$, so $\mathcal{T}' \subseteq \mathcal{T}$. For all $V \in \mathcal{T}$, there exists $B_x \in \mathcal{B}$ for all $x \in V$, then $\bigcup_{x \in V} B_x = V$, which implies $\mathcal{T} \subseteq \mathcal{T}'$. $\square$

Since the open set is a union of basis elements, the basis of a topology is not unique.

Proposition 3. Let X be a topological space. Suppose that $\mathcal{C}$ is a collection of open sets of X such that for each open set U of X and each $x \in U$, there is an element $C \in \mathcal{C}$ such that $x \in C \subseteq U$. Then $\mathcal{C}$ is a basis for X .

Proof. Let X be a topology on a set S . For all $s \in S$, we have $s \in S \subseteq X$, then there exists $C \in \mathcal{C}$ such that $s \in C$. If $x \in C_1 \cap C_2$, since $C_1, C_2 \in X$, $C_1 \cap C_2 \in X$, hence $\mathcal{C}$ is a basis. Now let $\mathcal{T}$ be the topology generated by $\mathcal{C}$, since elements in $\mathcal{T}$ can be written as union of basis elements and $C_i \in X$, $\mathcal{T} \subseteq X$. Pick $U \in X$, for all $x_i \in U$, there exists $C_i \in \mathcal{C}$ such that $x_i \in C_i \subseteq U$, which implies $U = \bigcup C_i \in \mathcal{T}$. Hence $X = \mathcal{T}$. $\square$

Proposition 4. Let $\mathcal{B}$ and $\mathcal{B}'$ be bases of topologies $\mathcal{T}$ and $\mathcal{T}'$ on X , respectively. Then the following are equivalent:

1. $\mathcal{T}'$ is finer than $\mathcal{T}$;
2. for each $x \in X$ and basis element $x \in B \in \mathcal{B}$, there exists a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subseteq B$.

Proof. $(\Rightarrow)$ Let $\mathcal{T} \subseteq \mathcal{T}'$. For all $x \in X$ and $x \in B \in \mathcal{B}$, since $\mathcal{T} \subseteq \mathcal{T}'$, $B \in \mathcal{B} \subseteq \mathcal{T} \subseteq \mathcal{T}'$, then there exists $x \in B' \subseteq B$. $(\Leftarrow)$ Fix $U \in \mathcal{T}$, for all $x \in U$ and for all $x \in B \in \mathcal{B}$, there exists $B' \in \mathcal{B}'$ such that $x \in B' \subseteq B \subseteq U$, so $U \in \mathcal{T}'$, which means $\mathcal{T} \subseteq \mathcal{T}'$. $\square$

Example. Bases of topologies

1. Any topology is a basis for itself.
2. The collection of open balls forms the basis of the standard topology on $\mathbb{R}^n$.

Vocabulary 1.9 (subbasis). A subbasis $\mathcal{S}$ for a topology on X is a collection of subsets of X whose union is X . The topology generated by $\mathcal{S}$ is the collection $\mathcal{T}$ of all unions of finite intersections of elements of $\mathcal{S}$.

Remark. Equivalently, a subbasis $\mathcal{S}$ induces a basis $\mathcal{B}$ that consists of finite intersections of elements in $\mathcal{S}$. For all $x \in X$, there exists $S \in \mathcal{S}$ such that $x \in S$. Let $x \in \bigcap_{i=1}^n S_i \cap \bigcap_{j=1}^m S_j$, where $S_i, S_j \in \mathcal{S}$, then $x \in \bigcap_{i,j} S_i \cap S_j$, a finite intersection. Therefore, a subbasis generates a basis.

Continuous Functions and Closed Sets

Discussed the relationship between continuous functions and closed sets in topological spaces. Defined homeomorphisms. Described the interior and closure of sets within topological spaces.

Remark. Let $f : A \rightarrow B$ be a function, $f^{-1}(b)$, where $b \in B$, is the preimage.

Vocabulary 2.1 (continuous function). Given two topological spaces X and Y . A continuous function $f : X \rightarrow Y$ is a function such that for all $V \in \mathcal{T}_Y$, $f^{-1}(V) \in \mathcal{T}_X$.

One can show this definition is equivalent to the $\epsilon - \delta$ definition in metric spaces.

Vocabulary 2.2 (homeomorphism). A continuous function is a homeomorphism if its inverse is also continuous.

Example. A **path** in a space X is a continuous function $\gamma : [0, 1] \rightarrow X$. A **loop** in a space X is a continuous function $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = \gamma(1)$. We shall see those definitions later.

Example. The interval $(0, 1)$ is homeomorphic to $\mathbb{R}$. Moreover, any open ball $B \subseteq \mathbb{R}^n$ is homeomorphic to $\mathbb{R}^n$.

Example. Not every continuous bijection between topological spaces is a homeomorphism. Consider the identity function $\text{id} : (\mathbb{R}, \mathcal{T}_1) \rightarrow (\mathbb{R}, \mathcal{T}_2)$, where $\mathcal{T}_1$ is the discrete topology and $\mathcal{T}_2$ is the standard topology.

Proof. It is trivial that id is continuous. However, consider the inverse of it, not every subset in $\mathcal{T}_2$ is open. $\square$

Vocabulary 2.3 (closed set of a topological space). Let $\mathcal{T}$ be a topological space on X . A set V is closed if $X \setminus V \in \mathcal{T}$. In other words, $V \subseteq X$ is closed if its complement $X \setminus V$ is open.

There is an equivalent way of stating continuity in terms of closed sets.

Proposition 5. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces. Let $f : X \rightarrow Y$ be a function, then f is continuous if and only if for any closed set $C \in Y$, $f^{-1}(C)$ is closed in X .

Proof. Let $C \subseteq Y$ be a closed subset, then $Y \setminus C$ is open and f is continuous implies $f^{-1}(Y \setminus C)$ is open in X , that is, $X \setminus f^{-1}(Y \setminus C) = f^{-1}(C)$ is closed in X . A Similar proof interchanging closed and open shows the converse. $\square$

There is another way to state this definition in terms of neighborhoods.

Proposition 6. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces, then f is continuous if and only if f is continuous at all $x \in X$, i.e. for each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subseteq V$.

Proof. Let f be a continuous function. Take a neighborhood V of $f(x) \in Y$. Then $x \in f^{-1}(V)$, and hence a neighborhood of x . Furthermore, $f(f^{-1}(U)) \subseteq V$. To show the converse, take any open set $V \subseteq Y$, and any $x \in f^{-1}(V)$, then $f(x) \in f(f^{-1}(V)) \subseteq V$. By continuity at x , there exists an open neighborhood U_x of x such that $f(U_x) \subseteq V$. Then $x \in U_x \subseteq f^{-1}(V)$ for any $x \in X$. Hence, $\bigcup_{x \in f^{-1}(V)} U_x = f^{-1}(V)$, then $f^{-1}(V)$ is open. $\square$

Proposition 7. The composition of continuous maps is continuous.

Proof. Let X, Y, Z be topological spaces, $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be continuous functions. Then $(gf)^{-1}(W) = (f^{-1}g^{-1})(W) = f^{-1}(g^{-1}(W))$. Since g is continuous, $g^{-1}(W)$ is open in Y . Since f is continuous, $f^{-1}(g^{-1}(W))$ is open in X . $\square$

Topological spaces with continuous functions (as morphisms) form a category **Top** and homeomorphisms are the isomorphisms. Please check the appendix for a more categorical approach.

Vocabulary 2.4 (interior and closure). Let X be a topological space and $A \subset X$. The interior of A , denoted $\text{Int}A$, is the union of all open sets contained in A . The closure of A , denoted $\overline{A}$, is the intersection of all closed sets containing A . Observe that $\text{Int}A$ is open and $\overline{A}$ is closed. Moreover, if A is open, $A = \text{Int}A$; if A is closed, $A = \overline{A}$.

Proposition 8. Let Y be a subspace of X , A be a subset of Y , and $\overline{A}$ denote the closure of A in X . Then the closure of A in Y is $\overline{A} \cap Y$.

Proposition 9. Let X be a topological space and $A \subseteq B \subseteq X$, then $\overline{A} \subseteq \overline{B}$.

Proof. If C is closed and $B \subseteq C$, then $A \subseteq B \subseteq C$, so the collection of closed sets containing A is a subcollection of closed sets containing B , hence $\overline{A} \subseteq \overline{B}$. $\square$

Proposition 10. Let A be a subset of the topological space X . Then the following statements are true:

1. $x \in \overline{A}$ if and only if every neighborhood of x intersects A ;

2. suppose the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element containing x intersects A .

Proof. (i) Let $\mathcal{T}$ be the topology on X and $A \subseteq X$. ($\Rightarrow$) Let U_x be a neighborhood of x and $U_x \cap A = \emptyset$, then $\bar{A} \subseteq X \setminus U_x$, so $x \notin \bar{A}$. ($\Leftarrow$) Let $x \notin \bar{A}$, since $\bar{A}$ is closed, $X \setminus \bar{A}$ is open. We have $x \in X \setminus \bar{A} \cap A$. (ii) This is an immediate consequence of (i). $\square$

Proposition 11. Let X and Y be topological spaces and $f : X \rightarrow Y$ be a function. Then f is continuous if and only if for any $A \subseteq X$, $f(\bar{A}) \subseteq \overline{f(A)}$.

Proof. Define $\bar{A} = \{x \in X : A \cap U \neq \emptyset\}$ and $\overline{f(A)} = \{y \in Y : V \cap f(A) \neq \emptyset\}$, where U and V are neighborhoods of x and y , respectively. For all $x \in \bar{A}$ and neighborhood V of $f(x)$, let $U = f^{-1}(V)$ and U is open and $x \in U$. Since $x \in \bar{A}$, then $A \cap U \neq \emptyset$. So there exists $x' \in A \cap f^{-1}(V)$. Then $f(x') \in f(A)$ and $f(x') \in f(f^{-1}(V)) \subseteq V$. Then $f(x') \in f(A) \cap V$, which implies $f(A) \cap V \neq \emptyset$, which implies $f(x) \in \overline{f(A)}$. $\square$

Vocabulary 2.5 (limit point). Let X be a topological space and $A \subset X$. A point $x \in X$ is a limit point of A if every neighborhood of x intersects A in some point other than x .

Proposition 12. Let A be a subset of a topological space X and A' be the set of all limit points of A , then $\bar{A} = A \cup A'$.

Proof. Let $x \in \bar{A}$ and $x \notin A$, then for all neighborhood U_x , $A \cap U_x \neq \emptyset$, so $x \in A' \subseteq A \cup A'$. It is trivial that $A \subseteq \bar{A}$. For all $x \in A'$ and neighborhood U_x of x , $U_x \cap A \neq \emptyset$, so $x \in \bar{A}$. Hence $\bar{A} = A \cup A'$. $\square$

Example. Consider the real line $\mathbb{R}$ and $A = (0, 1]$, $\bar{A} = [0, 1]$. Every point in $[0, 1]$ is a limit point of A .

Several Types of Topology

In this section, we construct topologies out of the given ones. We introduce subspace topology, quotient topology, product topology, disjoint union topology, and metric topology. Cell complex is built using these.

Subspace Topology

Vocabulary 3.1 (subspace topology). Let X be a set with topology $\mathcal{T}_X$. If $Y \subset X$, then the collection $\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}_X\}$ is a topology on Y , called the subspace topology, and Y is called a subspace of X .

Remark. It suffices to check that $\mathcal{T}_Y$ is a topology. Pick $\emptyset \in \mathcal{T}_X$, $Y \cap \emptyset = \emptyset$; then pick $X \in \mathcal{T}_X$, $Y \cap X = Y$. Consider the union $\bigcup_{i \in I} U_i$, where $U_i \in Y$, then there exists $V_i \in X$ such that $V_i \cap Y = U_i$, so $\bigcup_{i \in I} U_i = \bigcup_{i \in I} (V_i \cap Y) = Y \cap \bigcup_{i \in I} V_i$, where $\bigcup_{i \in I} V_i \in \mathcal{T}_X$, so $\bigcup_{i \in I} U_i \in \mathcal{T}_Y$. Similarly, $\bigcap_{i=1}^n U_i \in \mathcal{T}_Y$. Hence $\mathcal{T}_Y$ is a topology.

Proposition 13. If $\mathcal{B}$ is a basis for the topology of X , then the collection $\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$ is a basis for the subspace topology on Y .

Proof. For all $B \cap Y \in \mathcal{B}_Y$, since $\mathcal{B}$ generates $\mathcal{T}_X$, $B \in \mathcal{T}_X$, so $B \cap Y \in \mathcal{T}_Y$. For all $x \in U \cap Y$, where $U \in \mathcal{T}_X$, we can rewrite U as a union of $B_i \in \mathcal{B}$, and there exists a basis element B such that $x \in B$, so $x \in B \cap Y \subset U \cap Y$. Hence $\mathcal{B}_Y$ is the basis of $\mathcal{T}_Y$. $\square$

Proposition 14. Let Y be a subspace of X . If U is open in Y and Y is open in X , then U is open in X .

Proof. Since $U \in \mathcal{T}_Y$, then there exists $V \in \mathcal{T}_X$ such that $U = V \cap Y$. Since $Y \in \mathcal{T}_X$, $U = V \cap Y \in \mathcal{T}_X$. $\square$

Proposition 15. If $f : \mathcal{T}_X \rightarrow \mathcal{T}_Y$ is a continuous function, let $A \subseteq X$ be a subset with a subspace topology $\mathcal{T}_A$ of $\mathcal{T}_X$, then $f|_A : A \rightarrow Y$ is continuous.

Proof. Take any $V \in \mathcal{T}_Y$, $(f|_A)^{-1}(V) = A \cap f^{-1}(V)$. Since f is continuous, $f^{-1}(V) \in \mathcal{T}_X$ is open, hence the intersection $A \cap f^{-1}(V)$ is open in $\mathcal{T}_A$. $\square$

Example. Some examples of subspace topology

1. The subspace topology defined on the interval $[0, 1]$ as a subspace of $\mathbb{R}$ is both open and closed. As a subset of $\mathbb{R}$, or as an element of the standard topology $\mathcal{T}_{\mathbb{R}}$ on $\mathbb{R}$, $[0, 1]$ is only closed.

2. $S^1 \subseteq \mathbb{R}^2$ is a subspace topology. The open sets in S^1 are disjoint unions of curved open intervals. More generally, the n -dimensional sphere $S^n \subseteq \mathbb{R}^{n+1}$ is a subspace topology of $n+1$ -dimensional real coordinate space. The open sets are disjoint unions of n -dimensional curved open balls. Formally, any neighborhood $U \subseteq S^n$ of the point $x_0 \in S^n$ is homeomorphic to an open set in $\mathbb{R}^n$.
3. The subspace topology $\mathbb{Q} \subseteq \mathbb{R}$ is not the discrete topology on $\mathbb{Q}$.

Quotient Topology

Suppose $\pi : X \rightarrow S$ is a surjective map, π can be viewed as an equivalence relation on X such that for all $x, y \in X$, $x \sim y$ if and only if $\pi(x) = \pi(y)$.

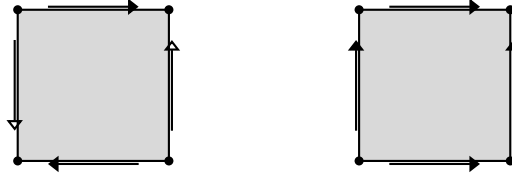
Vocabulary 3.2 (quotient topology). Let $(X, \mathcal{T})$ be a topological space and $f : X \rightarrow S$ be a surjective map. A set $U \subseteq S$ is open in the quotient topology if and only if $f^{-1}(U)$ is open in X .

Remark. It suffices to show this defines a topology $\mathcal{T}_Y$. Consider $f^{-1}(\emptyset) = \emptyset \in \mathcal{T}$, similarly, $f^{-1}(Y) = X \in \mathcal{T}$, so $Y, \emptyset \in \mathcal{T}_Y$. Consider $V_i \in \mathcal{T}_Y$, then $f^{-1}(V_i) \in \mathcal{T}$. Now consider $f^{-1}(\bigcup_{i \in I} V_i) = \bigcup_{i \in I} f^{-1}(V_i) \in \mathcal{T}_X$. Similarly, any finite intersection lies in the collection.

Vocabulary 3.3 (saturated subset). Let $f : X \rightarrow Y$ be a function between sets. A subset $U \subseteq X$ is saturated if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies either $x_1 \in U$ and $x_2 \in U$ or $x_1 \notin U$ and $x_2 \notin U$.

Example. The *projective space* $\mathbb{RP}^n$ is defined to be the quotient $\mathbb{R}^{n+1} \setminus \{0\}$ with the relation $x \sim \lambda x$ for $\lambda \in \mathbb{R}$.

Example. Consider the unit square I^2 and the topology T^2 defined by $I^2 / \sim$ with $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$, this is called the *torus*. We could construct the projective plane alternatively by unit square. The following two diagrams are the $\mathbb{RP}^2$ and torus, respectively.



Proof. I will give a sketch on the equivalent definitions of $\mathbb{RP}^2$. (1) replace the unit square I^2 by the unit disk D^2 (2) identify the disk D^2 with the upper hemisphere H of the unit sphere S^2 (3) inclusion map from H to S^2 induces a homeomorphism from $H / \sim$ to $S^2 / \sim$, where the equivalence relation $\sim$ identifies antipodal points (4) construct a homeomorphism between $S^2 / \sim$ and $\mathbb{RP}^2$ $\square$

Product Topology

Vocabulary 3.4 (finite product topology). Let X and Y be topological spaces. The finite product topology on $X \times Y$ is the topology having as basis the collection $\mathcal{B}$ of all sets of the form $U \times V$, where U is an open subset of X and V is an open subset of Y .

Now we check $\mathcal{B}$ is a basis. For all $(x, y) \in X \times Y$, since $X \in \mathcal{T}_X$ and $Y \in \mathcal{T}_Y$, $(x, y) \in X \times Y$. If $(x, y) \in (A \times B) \cap (C \times D)$, then $x \in A \cap C$ and $y \in B \cap D$, so there exists $(x, y) \in (A \cap C) \times (B \cap D)$. Hence $\mathcal{B}$ is a basis.

Proposition 16. If $\mathcal{B}$ is a basis for the topology on X and $\mathcal{C}$ is a basis for the topology on Y , then $\mathcal{D} = \{B \times C \mid B \in \mathcal{B} \text{ and } C \in \mathcal{C}\}$ is a basis for the topology of $X \times Y$.

Proof. Let $W \subset X \times Y$ and $(x, y) \in W$, then there exists $U \times V$ such that $(x, y) \in U \times V \subset W$. Since $\mathcal{B}$ and $\mathcal{C}$ are bases of X and Y , respectively, there exists $B \in \mathcal{B}$ and $C \in \mathcal{C}$ such that $x \in B \subset U$ and $y \in C \subset V$, then $(x, y) \in B \times C \subset W$. Hence $\mathcal{D}$ is a basis for the product topology. $\square$

Vocabulary 3.5. Let $\pi_1 : X \times Y \rightarrow X$ and $\pi_2 : X \times Y \rightarrow Y$ be functions such that $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$. The maps π_1 and π_2 are called the *projections* of $X \times Y$ onto its first and second factors, respectively.

Proposition 17. Let $X_1 \times X_2$ be a product topology, then the projections π_1 and π_2 are continuous.

Proof. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be the topologies on X_1 and X_2 , respectively. Let $U_1 \in \mathcal{T}_1$,

$$\pi^{-1}(U_1) = \{(x_1, x_2) \in X_1 \times X_2 \mid x_1 \in U_1\} = U_1 \times X_2 \subseteq X_1 \times X_2$$

Since $U_1 \in \mathcal{T}_1$ and $X_2 \in \mathcal{T}_2$, $\pi^{-1}(U_1) \in \mathcal{B} \subseteq \mathcal{T}(\mathcal{B})$. $\square$

Example. The *torus* T^2 can be rewrite by $T^2 = S^1 \times S^1$. One could verify the two constructions yield the same topology up to homeomorphism.

Proposition 18. Consider any function $f : Y \rightarrow X_1 \times X_2$, let $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$, then f is continuous if and only if f_1 and f_2 are continuous.

Proof. ($\Rightarrow$) For all basis elements $U_1 \times U_2$, $f^{-1}(U_1 \times U_2) = \{y \in Y \mid f(y) \in U_1 \times U_2\} = \{y \in Y \mid (f_1(y), f_2(y)) \in U_1 \times U_2\} = \{y \in Y \mid f_1(y) \in U_1, f_2(y) \in U_2\} = \{y \in Y \mid y \in f_1^{-1}(U_1), y \in f_2^{-1}(U_2)\} = f_1^{-1}(U_1) \cap f_2^{-1}(U_2)$. Since f_1 and f_2 are continuous, $f_1^{-1}(U_1)$ and $f_2^{-1}(U_2)$ are open in Y , hence $f^{-1}(U_1 \times U_2)$ is open in Y . ($\Leftarrow$) The composition of continuous functions is continuous. $\square$

Vocabulary 3.6 (coordinate). Let J be an indexed set. Given any set X , a J -tuple of elements of X to be a function $x : J \rightarrow X$. If $\alpha \in J$, denote the value of x at α by x_α , this is called the α th coordinate.

Vocabulary 3.7 (cartesian product). Let $\{A_\alpha\}_{\alpha \in J}$ be an indexed family of sets. Let $X = \bigcup_{\alpha \in J} A_\alpha$. The cartesian product, denoted $\prod_{\alpha \in J} A_\alpha$, is defined to be the set of all J -tuples $(x_\alpha)_{\alpha \in J}$ of X such that $x_\alpha \in A_\alpha$ for each $\alpha \in J$.

Suppose $\{(X_i, \mathcal{T}_i)\}$ is a family of topological spaces. Take any $j \in I$, let $\pi_j : \prod_{i \in I} X_i \rightarrow X_j$ by $\pi_j(x) = x_j$. The preimage $\pi_j^{-1}(U_j)$ should be open for any $U_j \in \mathcal{T}_j$ for all $j \in I$. Take the collection of all such preimages, denotes $\mathcal{S} = \{\pi_j^{-1}(U_j) \mid j \in I, U_j \in \mathcal{T}_j\}$, $\mathcal{S}$ is not closed under the intersection. However, it suffices to check $\mathcal{S}$ is a subbasis, which induces a basis $\mathcal{B}(\mathcal{S})$ through finite intersections of sets of the form $\pi_j^{-1}(U_j)$.

Vocabulary 3.8 (general product topology). Let $\{X_\alpha\}_{\alpha \in A}$ be an arbitrary collection of topological spaces. The general product topology on X is defined to be the topology generated by the basis

$$\left\{ \prod_{\alpha \in A} U_\alpha \mid U_\alpha \subseteq X_\alpha \text{ is open and all but finitely many } U_\alpha = X_\alpha \right\}$$

Disjoint Union Topology

Let $\{X_i, \mathcal{T}_i\}_{i \in I}$ be a family of topological spaces. Taking the disjoint union of $\bigsqcup_{i \in I} X_i = \{X_i \times \{i\}\}_{i \in I}$ by taking copies of X_i disjointedly, represented by taking a different index for each copy.

Vocabulary 3.9. Let $\{X_\alpha\}_{\alpha \in A}$ be an arbitrary collection of topological spaces. The *disjoint union topology* is the topology generated by $\bigsqcup_{i \in I} \mathcal{T}_i$.

In the category **Top**, disjoint union is exactly the same as coproduct. Please check the appendix for a precise definition of the coproduct topology and examples.

Vocabulary 3.10 (dense). Let $(X, \mathcal{T})$ be a topological space, a nonempty subset $D \subset X$ is dense if for all $U \in \mathcal{T}$ and $U \neq \emptyset$, $D \cap U \neq \emptyset$.

Vocabulary 3.11 (separable). A topological space is separable if it admits a countable dense subset.

Cell Complex

We want to develop a topological construction of the combinatorial graph. A graph is consisting of a collection of vertices and edges. Vertices is the topological equivalence of disc of dimension zero D^0 . Edges are the same as D^1 , one dimensional disc. The construction of a graph is by identifying edges with the vertices they are attached to. This gluing action correspond to forming an equivalence relation on the disjoint union of copies of D^0 union with copies of D^1 .

Notice that equivalence relation is transitive. Hence if the endpoint of two edges are attached to the same vertices, the equivalence class will consist of three points. Similar idea if more points are attached.

Vocabulary 3.12 (cell complex). Graph is a special example of the more general construction of cell complex (also known as CW complex). In general, a cell complex is constructed in these steps:

- I Take the disjoint union of several copies of D^0 . Take the quotient topology by the empty equivalent relation. This is equal to not doing anything. More formally, the 0-skeleton is defined as $X^0 = \coprod D^0$
- II Take the disjoint union of the 0-skeleton X^0 with several copies of D^1 . Then identify how to glue the boundaries (endpoints) of the 1-discs to the 0-discs. Such action correspond to the equivalent relation and modded out the disjoint union by this equivalence relation. That is, $X^1 = (X^0 \sqcup (\coprod D^1)) / \sim$
- III Apply this procedure recursively. Define the n -skeleton as the disjoint union of $n - 1$ -skeleton with copies of D^n . Then identify the equivalence relation from the boundary of n -discs to $n - 1$ discs and modded out by this. Formally,

$$X^n = \left(X^{n-1} \sqcup \left(\coprod D^n \right) \right) / \sim$$

- IV This procedure can either go on for finitely many steps, or countably infinitely many steps. One can consider the cell complex of finite dimension n as the cell complex attaching no copies of D^m for $m \geq n + 1$.
- V The ∞ -skeleton of this cell complex construction is defined as taking the colimit of the chains. Essentially, colimit captures the amalgamation of a chain of maps (and limit showcase intersection of the chain). We will not discuss what colimit is here formally. That requires extensive discussion on category theory and some homological algebra.

Remark. Equivalently, there is a similar, but more structured description of cell complexes (that can be easily generalized to other complex structures). Let X^0 be a discrete set (consisting of disjoint points). From the n -skeleton X^n , we attach a family of e_α^n of n -cells along the attaching maps $\phi_\alpha : S^{n-1} \rightarrow X^{n-1}$. These data can be summarized as the following diagram

$$\begin{array}{ccc} \coprod_{\alpha \in A_n} S^{n-1} & \longrightarrow & \coprod_{\alpha \in A_n} D^n \\ (\phi_\alpha) \downarrow & & \downarrow (\Phi_\alpha) \\ X^{n-1} & \longrightarrow & X^n \end{array}$$

via $X^n = X^{n-1} \sqcup (\coprod_{\alpha \in A_n} D^n) / \sim$. Take $X = \bigcup_{n \in \mathbb{N}} X^n$ and topologize the quotient $\coprod X^n \rightarrow X$. Note that if $n < \infty$, we take $X = X_n$ and this topologization is redundant.

We call $\Phi_\alpha : D^n \rightarrow X^n \subseteq X$ the characteristic map. $X = \coprod \Phi_\alpha(\text{int} D^n)$ is a quotient $(\Phi_\alpha) : \coprod_{n,\alpha} D^n \rightarrow X$. Its restrictions to S^{n-1} , $\phi_\alpha = \Phi|_{S^{n-1}} : S^{n-1} \rightarrow X^{n-1} \subseteq X^n$ are attaching maps. Then (ϕ_α) determines the gluing action and indicate how the complexes are constructed. Check the appendix for its significance in algebraic topology.

Metric Topology and Convergence

We first introduce the basic notions of norms followed by some results related to them. Then we described what it means to have a sequence in a space. And with all that setup, we can finally talk about convergence.

Vocabulary 4.1 (norm). A norm on a vector space V over $\mathbb{C}$ is a function $\|\cdot\| : V \rightarrow \mathbb{C}$ satisfies:

1. $\|v\| \geq 0$ for all $v \in V$, $\|v\| = 0$ if and only if $v = 0$;
2. $\|v + w\| \leq \|v\| + \|w\|$ for all $v, w \in V$;
3. $\|\alpha v\| = |\alpha| \|v\|$ for all $v \in V$ and $\alpha \in \mathbb{C}$.

Vocabulary 4.2 (p -norm). For all $p \geq 1$, the p -norm on $\mathbb{R}^n$ is defined by $\ell^p((x_n)) := \|(x_1, \dots, x_n)\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$.

Vocabulary 4.3 (supremum norm). Define $d_{\text{sup}}(f, g) = \sup\{d_Y(f(x), g(x)) \mid x \in X\}$. This is not a norm as the supremum could be infinite. To tackle this, we need to choose f and g from the set of bounded functions.

Vocabulary 4.4 (infinity norm). Let us consider a special case of the supremum norm. When $Y = \mathbb{R}$, $d_{\text{sup}}(f, g) = \sup\{\|(f(x), g(x))\| \mid x \in X\}$, induced by the norm $\|f\|_{\text{sup}} = \sup\{\|f(x)\| \mid x \in X\}$. Denote $\ell^\infty(X, \mathbb{R}) = \{f : X \rightarrow \mathbb{R} \mid \|f\|_{\text{sup}} < \infty\}$.

Proposition 19. Let X be a metric space, suppose there exists an uncountable subset $A \subset X$ and a constant $C > 0$ such that $d(a, b) > C$ for distinct $a, b \in A$, then A is not separable.

Proof. For all $a \in A$, let $U_a = B(a, C/2)$. Let $D \subset X$ be dense and define $D_a = D \cap U_a$, then $D_a \neq \emptyset$. All U_a and D_a are pairwise disjoint. The sets D_a form a partition for $D' = \bigcup_{a \in A} D_a \subset D$, then there exists $f : D' \rightarrow A$ with $x \mapsto a$, where $x \in D'$ and $x \in D_a$. It follows that $|A| \leq |D'| \leq |D|$. Since A is uncountable, D is uncountable. $\square$

Every normed vector space is a metric space, which induces a topological space. Given a norm on V over $\mathbb{C}$, for all $x, y, z \in V$, define $d(x, y) = \|x - y\|$, so $d(x, y) \geq 0$ and the norm $\|x - y\|$ is 0 if and only if $x = y$. Also, $d(x, y) = \|x - y\| = |-1|\|y - x\| = \|y - x\|$ and $\|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\|$, so a norm induces a metric space. We show that all open sets in a metric space (X, d) form a topology. Let the family of all open sets in X , denoted $\mathcal{T}$. Fix $x_0 \in X$, then the open ball $B(x_0, 0) = \emptyset$. There exists an open set consists with all open balls in X , which is exactly X . Let U_1 and U_2 be open sets, we can rewrite them by union of open balls, and the union of open balls is again an open set, similarly, the finite intersection is still an open set, so (X, d) induces a topology. Hence every norm induces a topology.

Example. Consider the set of all sequences in $\mathbb{R}$ without divergent norm, denotes this set by $\mathbb{R}^{\mathbb{N}}$. The set ℓ^p of sequences $\{x_n\}$ for which $\sum_{n=1}^{\infty} x_n^p$ is finite with $\|x_i^p\|_p = (\sum_{n=1}^{\infty} |x_i|^p)^{1/p}$ is a normed vector space. Similarly, we define ℓ^∞ to be the set of bounded sequences with $\|x_i^p\|_\infty = \sup\{|x_i|\}$ is a normed vector space. For all finite $p, q \in \mathbb{N}$, $\ell^p \simeq \ell^q$ and $\ell^p \neq \ell^\infty$.

Proof. Given $\ell^p \simeq \ell^q$, we want to prove $\ell^p \neq \ell^\infty$. Consider ℓ^p , define $D = (x_n)_n$ such that for all $n, x_n \in \mathbb{Q}$, there are finitely many nonzero x_n . Then $D = \mathbb{Q} \times \cdots \times \mathbb{Q} \times \{0\} \times \cdots$ is a countable union of countable sets. It follows that D is countable. Let x be a sequence in ℓ^p and let $\varepsilon > 0$. For all $m \geq 0$, define a sequence

$$x^m = \begin{cases} x_n^m = x_n & \text{if } n \leq m \\ x_n = 0 & \text{if } n > m \end{cases}$$

$\sum_{n=m+1}^{\infty} |x_n|^p$ converges to 0 implies that there exists a sufficiently large m such that $\|x - x^m\|_p \leq \varepsilon/2$. Approximate each $x_i \in \mathbb{R}$ by rationals y_i such that $|x_i - y_i|^p < \varepsilon/(2(m+1))$, then $\|x^m - y\|_p < \varepsilon/2$, which implies $\|x - y\|_p < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Hence D is dense and ℓ^p is separable. Now we consider the ℓ^∞ case. For all subset A , define the sequence a such that $a_n = 1$ if $n \in A$ and $a_n = 0$ otherwise. Given two distinct subsets $A, B \subset \mathbb{N}$, then there exists $n \in \mathbb{N}$ such that either $n \in A$ and $n \notin B$ or $n \notin A$ and $n \in B$, which implies $|a_n - b_n| = 1$, so $\|a_n - b_n\|_\infty = 1$. The set $\{a \mid A \subset \mathbb{N}\}$ is uncountable. By proposition 19, since set P exists, ℓ^∞ is not separable. Hence $\ell^p \neq \ell^\infty$. $\square$

Example. Consider the harmonic series $x_i = i^{-1}$ for $i \in \mathbb{N}$. We know that $\sum_1^\infty |i^{-1}| = \sum_1^\infty i^{-1}$ diverges, while $\sum_1^\infty |i|^2 = \sum_1^\infty i^{-2}$ converges. Hence, this is not a equality as the harmonic series belongs to ℓ_2 but not ℓ_1 .

Vocabulary 4.5 (metrizable). A topological space $(X, \mathcal{T})$ is metrizable if there exists a metric d on X such that $\mathcal{T}$ is generated by the family of balls in X with respect to d .

Vocabulary 4.6 (sequence). A sequence in a topological space X is a function $x : \mathbb{N} \rightarrow X$ where $i \mapsto x_i$, denoted by $(x_i)_{i \in \mathbb{N}}$. A sequence (x_i) in X converges to $x \in X$, called a limit point of $(x_i)_{i \in \mathbb{N}}$, if for all neighborhoods U of x , there exists $N \in \mathbb{N}$ such that for all $i \geq N$, $x_i \in U$.

Proposition 20. Let X be a topological space, $A \subseteq X$ a subset, and (x_n) be a sequence in A converges to some $x \in X$. Then $x \in \overline{A}$. Conversely, if X is metrizable, then for all $x \in \overline{A}$, there exists a sequence (x_n) converges to x .

Proof. Given $x \in X$, pick a neighborhood U of x in X . Let (x_i) be a sequence in A that converges to x , then there exists $N \in \mathbb{N}$ such that, for all $i \geq N$, $x_i \in U$, so $x \in A \cap U$, which implies $A \cap U \neq \emptyset$. Hence $x \in \overline{A}$. Conversely, let X be metrizable and let $x \in \overline{A}$. Consider the sequence of balls $B(x, i^{-1})$ for $i \in \mathbb{N}$ with respect to the metric d that generates the topology. For all i , $A \cap B(x, i^{-1}) \neq \emptyset$, so there exists a point $x_i \in A \cap B(x, i^{-1})$. This gives a sequence (x_i) in A . Pick any neighborhood U of x in X , since the topology is generated by a metric, there exists $r > 0$ such that $B = B(x, r) \subseteq U$, let N be sufficiently large so that $B(x, N^{-1}) \subseteq B$, so for all $i \geq N$, $x_i \in B(x, i^{-1}) \subseteq B(x, N^{-1}) \subseteq B \subseteq U$. $\square$

Vocabulary 4.7 (first-countable). A topological space X is first-countable if each point has a countable neighborhood basis.

Vocabulary 4.8 (second-countable). A topological space X is second-countable if X admits a countable basis.

Remark. Second-countable is stronger than first-countable (obviously).

Vocabulary 4.9 (Lipschitz equivalent). If two metrics d and d' are given on a set X , we say d and d' are Lipschitz equivalent if there exists $c_1, c_2 > 0$ such that for all $x, y \in X$, $c_1 d(x, y) \leq d'(x, y) \leq c_2 d(x, y)$.

Vocabulary 4.10 (Lipschitz). Let X and Y be two metric spaces, we say $f : X \rightarrow Y$ is Lipschitz if there exists $L > 0$ such that for all $x, x' \in X$, $d_Y(f(x), f(x')) \leq L \cdot d_X(x, x')$.

Proposition 21. Let $f : X \rightarrow Y$ be a function where X and Y are topological spaces. If f is continuous, then for any sequence (x_i) in X converging to $x \in X$, $(f(x_i))$ converges to $f(x)$ in Y . If X is metrizable, then for any sequence (x_i) converging to $x \in X$, we have $(f(x_i))$ converges to $f(x)$ in Y , then f is continuous.

Proof. Let f be continuous and $x_i \rightarrow x$ in X . Take any neighborhood V of $f(x)$ in Y , since f is continuous, $f^{-1}(V)$ is open in X . Since $x \in f^{-1}(V)$, it is a neighborhood, then there exists $N \in \mathbb{N}$ such that for all $i \geq N$, $x_i \in f^{-1}(V)$, so $f(x_i) \in f(f^{-1}(V)) \subseteq V$ for any $i \geq N$, then $f(x_i) \rightarrow f(x)$. Pick any point in $f(\bar{A})$, then it is of the form $f(x)$ for some $x \in A$. Since X is metrizable, there exists a sequence (x_i) in A such that $x_i \rightarrow x$ and $f(x_i) \rightarrow f(x)$, then $f(x) \in f(\bar{A})$. $\square$

Here is an alternate proof for the converse direction. Take any closed set $C \subseteq Y$, we have $\bar{C} = C$. Then $f^{-1}(C) \subseteq \overline{f^{-1}(C)}$. Pick $x \in \overline{f^{-1}(C)}$. Approximate this point by a sequence in $f^{-1}(C)$, say $(x_i) \rightarrow x$, then $f(x_i) \in f(f^{-1}(C)) \subseteq C$. Since $f(x_i) \rightarrow f(x)$, $f(x) \in \bar{C} = C$. This implies that $x \in f^{-1}(C)$, then $f^{-1}(C) = \overline{f^{-1}(C)}$, so $f^{-1}(C)$ is closed.

Vocabulary 4.11 (point-wise convergence). Given a family of functions $f_n : X \rightarrow Y$ where Y is a topological space. We say f_n converges point-wise to f if for all $x \in X$, $f_n(x) \rightarrow f(x)$. Explicitly, for any $x \in X$, and any neighborhood U of $f(x)$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $f_n(x) \in U$.

Vocabulary 4.12 (uniform convergence). We say $f_n \rightarrow f$ uniformly if for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $x \in X$, $f_n(x) \in B_\epsilon(f(x))$. For this to make sense, Y should be a metric space, and X is any set.

Lemma. Given a sequence $f_n : X \rightarrow \mathbb{R}$ and $f : X \rightarrow \mathbb{R}$, where X is any set, then $f_n \rightarrow f$ uniformly if and only if $\|f_n - f\|_{\sup} \rightarrow 0$. Similarly for $f, f_n : X \rightarrow Y$ where Y is a metric space. $f_n \rightarrow f$ uniformly if and only if $d_{\sup}(f_n, f) \rightarrow 0$.

Remark. Note that uniform convergence implies convergence, but the converse is not true. Consider the example where $f_n : [0, 1] \rightarrow \mathbb{R}$, $f_n(x) = x^n$. It is clear that f_n converges to f point-wise where

$$f(x) = \begin{cases} 0 & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$$

To show f_n does not converge uniformly, we need a theorem.

Theorem 1. If $f_n, f : X \rightarrow Y$, Y is a metric space, X is a topological space. f_n is continuous for each n , $f_n \rightarrow f$ uniformly, then f is continuous.

Proof. Take any $x_0 \in X$, take any neighborhood V of $f(x_0)$. First choose $\epsilon > 0$ small enough so that $B_\epsilon(f(x_0)) \subseteq V$. Since $f_n \rightarrow f$ uniformly, there exists $N \in \mathbb{N}$ so that for all $n \geq N$ and for all $x_0 \in X$, $f_n(x_0) \in B_{\epsilon/3}(f(x_0))$. In particular, for all $x_0 \in X$, $f_N(x_0) \in B_{\epsilon/3}(f(x_0))$. Since f_N is continuous, there exists a neighborhood U of x_0 such that $f_N(U) \subseteq B_{\epsilon/3}(f(x_0))$. So for all $x \in U$, $d(f(x_0), f(x))$ $\square$

Connectedness and Compactness

Went over definitions and properties of connected and path-connected spaces. Discussed the intermediate value theorem and uniform continuity on compact spaces. Lebesgue's number lemma is explained as a crucial concept for proving compactness through open coverings.

Vocabulary 5.1 (separation). Let X be a topological space, $A \subseteq X$. A separation of A is a pair of open sets $U, V \subseteq A$ such that $U \cap V = \emptyset$, $U \cup V = A$, where $U, V \neq \emptyset$. This implies that U and V are closed in A .

Vocabulary 5.2 (connected). A set is called connected if it does not admit a separation.

Vocabulary 5.3 (path). A path in a space X is any continuous function $p : [0, 1] \rightarrow X$. If $p(0) = a$ and $p(1) = b$, then p is a path from a to b .

Vocabulary 5.4 (path-connected). A space is path-connected if for all $a, b \in X$, there exists a path p from a to b .

Lemma. The closed interval $[0, 1]$ is connected.

Proof. Suppose for contradiction that $[0, 1]$ can be separated into two nonempty disjoint open sets A and B . Without loss of generality, assume $1 \in B$. Define $c = \sup A$. We will examine the possible placements of c to show a contradiction in any case.

- c cannot be in A : Since A is open, if c were in A , there would exist an $\epsilon > 0$ such that the interval $(c - \epsilon, c + \epsilon)$ is contained in A , contradicting c being the supremum (as $c + \epsilon/2$ would be a greater element still in A).
- c cannot be in B : Since B is open and disjoint from A , if c were in B , there would exist a $\delta > 0$ such that $(c - \delta, c + \delta)$ is entirely in B . This implies there are elements less than c (specifically, $c - \delta/2$) which are not upper bounds for A , contradicting the definition of c as the supremum of A .
- c cannot be equal to 0: If $c = 0$, this would imply A is empty as there are no positive elements less than 0 in $[0, 1]$, contradicting our assumption that A is nonempty.

Therefore, c is not in A nor B . This is a contradiction. $\square$

Proposition 22. If $A \subseteq X$ is path-connected, then it is connected.

Proof. Suppose not, then there exists a separation $X = U \cup V$ of X . Pick $a \in U$ and $b \in V$ using the fact that $U, V \neq \emptyset$. X is path-connected implies there exists a path p from a to b . Consider $p^{-1}(U)$, $p^{-1}(U) \subseteq [0, 1]$, which are both open. We know that $0 \in p^{-1}(U)$ and $1 \in p^{-1}(V)$. For X to be disconnected, $p^{-1}(U) \cap p^{-1}(V) = \emptyset$ and $p^{-1}(U) \cup p^{-1}(V) = [0, 1]$. Then there exists a separation of $[0, 1]$, which is impossible. Hence X is connected. $\square$

Let X be a topological space, we want to define an equivalence relation $\sim$ on X . For $x, y \in X$, $x \sim y$ if and only if there exists a connected subset of A of x containing both x and y .

Theorem 2 (intermediate value theorem). Let $f : X \rightarrow \mathbb{R}$ be a continuous function between a connected topological space X to $\mathbb{R}$ with standard topology. Let $a, b \in X$, and r is in between $f(a)$ and $f(b)$, then there exists $x \in X$ such that $f(x) = r$.

Proof. Assume that $f(a) \leq r \leq f(b)$, and suppose not for contradiction. That means for all $x \in X$, $f(x) \neq r$. Then $f^{-1}((-\infty, r)) \cup f^{-1}((r, \infty)) = X$ and $f^{-1}((-\infty, r)) \cap f^{-1}((r, \infty)) = \emptyset$.

Now let $f(a) \leq r \leq f(b)$, if $r = f(a)$ or $r = f(b)$, then take $x = a$ or $x = b$. Hence, we can assume that $f(a) < r < f(b)$, which implies $a \in f^{-1}((-\infty, r))$, $b \in f^{-1}((r, \infty))$. These sets are nonempty and open. This gives a separation of X , which is a contradiction. $\square$

Lemma. The image of any connected set under any continuous function is connected.

Proof. $f : X \rightarrow Y$ continuous, $A \subseteq X$ connected. Consider the restriction $f_A : A \rightarrow Y$. Check that $f|_A$ is continuous. Let $V \subseteq Y$ is an open subset, $f|_A^{-1}(V) = f^{-1}(V) \cap A$ is open in A , so $f|_A$ is continuous. A is connected implies $f(A) = Y$. Suppose $f(A)$ is not connected, then there is a separation $f(A) = V_1 \cup V_2$ in $f(A)$, meaning that there exists V'_1 open in X and V'_2 open in X such that $V_1 = f(A) \cap V'_1$ and $V_2 = f(A) \cap V'_2$. This means that $f^{-1}(V_1) = f^{-1}(V'_1)$ open in X and $f^{-1}(V_2) = f^{-1}(V'_2)$ open in X . V_1 and V_2 nonempty implies there is a separation of X , which is a contradiction. $\square$

Let X be a topological space, define $\sim$ as the connected component, and $\sim_p$ as the path connected component. That means, for all $x, y \in X$, $x \sim y$ if and only if there exists a path from x to y .

Lemma. If $x \sim_p y$, then $x \sim y$.

Proof. There exists a path $p : [0, 1] \rightarrow X$ such that $p(0) = x$ and $p(1) = y$. $[0, 1]$ is connected implies $p([0, 1]) \subseteq X$ is connected, which means $x \sim y$. $\square$

It is clear that $\sim_p$ and $\sim$ are equivalence relations on X .

Theorem 3. Any closed interval $[a, b]$ in $\mathbb{R}$ is compact.

proof idea. First use the fact that A is an open cover of $[a, b]$. Then consider the set

$$S = \left\{ c \in [a, b] \mid \exists A' \subseteq A \text{ such that } [a, b] \subseteq \bigcup A' \right\}$$

Let $a \in S$, $\bar{c} := \sup(S)$. Finally, we prove that $\bar{c} \in S$. □

Vocabulary 5.5 (compact). Suppose X is a topological space, $Y \subseteq X$, then Y is compact (in the subspace topology on Y). Y is compact if and only if for all families A of open subsets in X , if $Y \subseteq \bigcup A$, then there exists a finite subfamily A' of A such that $Y \subseteq \bigcup A'$.

Vocabulary 5.6 (locally compact). A topological space X is locally compact if for all $x \in X$, there exists a neighborhood U of x that is contained in some compact subset of X .

Lemma. Let X be a compact space, then any closed subset of X is compact. (closed in X , but compact is an absolute concept)

Proof. Take any closed subset $C \subseteq X$, let A be any open cover of C in X (these open sets in the cover are open in X), and $C \subseteq \bigcup A$. Take $A \cup \{X \setminus C\}$, if it is an open cover of X , X is compact. Then there exists a finite subcover $A' \subseteq A \cup \{X \setminus C\}$ of X . Then A' is also an open cover of C in X with respect to X . (each element open in X). Remove $X \setminus C$ from A' if needed. The rest is still a cover of C . Because $(X \setminus C) \cap C = \emptyset$. □

Lemma. Let X be a Hausdorff topological space, C is a compact subset of X . Then C is closed in X .

Proof. To prove something is closed, we prove its complement is open. Let Y be a compact subspace of the Hausdorff space X , $x_0 \in X - Y$. For each point of $y \in Y$, choose disjoint neighborhoods U_y and V_y of those points using the Hausdorff property. $\{V_y \mid y \in Y\}$ is a covering for Y in X . Thus, the set is finite: $V_{y_1}, \dots, V_{y_n}$. Then these two open sets are disjoint

$$U = U_{y_1} \cap \dots \cap U_{y_n} \quad V = V_{y_1} \cup \dots \cup V_{y_n}$$

by pair-wise disjointness. Suppose $z \in V$, then $z \in V_{y_i}$ for some i , but $z \notin U_{y_i}$ so $z \notin U$. Then U is a neighborhood of x_0 disjoint from Y proving the complement open. □

Theorem 4. The image of a compact space under a continuous map is compact.

Proof. Let $f : X \rightarrow Y$ be continuous; let X be compact. Let $\mathcal{A}$ be a covering of the set $f(X)$ by sets open in Y . The collection $\{f^{-1}(\mathcal{A}) \mid \mathcal{A} \in \mathcal{A}\}$ is a collection of sets covering X ; these sets are open in X because f is continuous. Hence finitely many of them, say $f^{-1}(\mathcal{A}_\infty), \dots, f^{-1}(\mathcal{A}_n)$, cover X . Then the sets $\mathcal{A}_\infty, \dots, \mathcal{A}_n$ cover $f(X)$. □

Theorem 5. The product of finitely many compact spaces is compact.

Proof. Main idea: show this is true for 2 spaces: $X \times Y$.

Step 1: Suppose we have two spaces X and Y with Y compact. Pick $x_0 \in X$ and N an open set of $X \times Y$ containing the slice $x_0 \times Y \subseteq X \times Y$. Now we should prove the existence of the tube: $W \times Y$, where W is the neighborhood of x_0 in X such that N contains the entire set. We have $U_i \times V_i$ as basis elements of $X \times Y$ that cover $x_0 \times Y$ in N . Since $x_0 \times Y$ is homeomorphic to Y , it is compact. Then there exists

$$U_1 \times V_1, \dots, U_n \times V_n$$

as a finite subcover. Can assume each of the basis element intersects $x_0 \times Y$. With this assumption, we can define $W = U_1 \cap \dots \cap U_n$ is an open set that contains x_0 . Let $x \times y$ be a point of $W \times Y$. Consider the point $x_0 \times y$ of the slice $x_0 \times Y$ having the same y -coordinate as this point. Now $x_0 \times y$ belongs to $U_i \times V_i$ for some i , so that $y \in V_i$. But $x \in U_j$ for every j (because x is in the intersection). Therefore, $x \times y \in U_i \times V_i$, and $\{U_i \times V_i\}$ covers the tube $W \times Y$. $\{U_i \times V_i\}$ lie in N , and since they cover $W \times Y$, the tube $W \times Y$ lies in N also.

Step 2: Let X and Y be compact spaces. Let C be an open covering of $X \times Y$. Given $x_0 \in X$, the slice $x_0 \times Y$ is compact and have the finite subcover $C' = \{A_1, \dots, A_m\}$ of C . Their union $N = A_1 \cup \dots \cup A_m$ is an open set containing $x_0 \times Y$; by Step 1, the open set N contains a $x_0 \times Y$ tube $W \times Y$, where W is open. C' also covers the tube. Thus, for each x in X , we can choose a neighborhood W_x of x such that the tube $W_x \times Y$ can be covered by finitely many elements of C . Since X is compact, we only have finitely many tubes to be finitely covered. Thus, $X \times Y$ can be finitely covered. □

Theorem 6. Every closed interval in $\mathbb{R}$ is compact.

Lemma. If $A \subseteq \mathbb{R}^n$, then A is compact if and only if A is bounded and closed in $\mathbb{R}^n$.

Proof. It will suffice to consider only the metric ρ ; the inequalities

$$\rho(x, y) \leq d(x, y) \leq \sqrt{n}\rho(x, y)$$

imply that A is bounded under d if and only if it is bounded under ρ . Suppose that A is compact. Then since compact subset of a Hausdorff space is closed, A is closed. Consider the collection of open sets $\{B_\rho(\mathbf{0}, m) | m \in \mathbb{Z}_+\}$ whose union is all of $\mathbb{R}^n$. Some finite subcollection covers A . It follows that $A \subset B_\rho(\mathbf{0}, M)$ for some M . Therefore, for any two points x and y of A , we have $\rho(x, y) \leq 2M$. Thus A is bounded under ρ .

Conversely, suppose that A is closed and bounded under ρ ; suppose that $\rho(x, y) \leq N$ for every pair x, y of points of A . Choose a point x_0 of A , and let $\rho(x_0, \mathbf{0}) = b$. The triangle inequality implies that $\rho(x_0, \mathbf{0}) \leq N + b$ for every x in A . If $P = N + b$, then A is a subset of the cube $[-P, P]^n$, which is compact. Being closed, A is also compact. $\square$

Theorem 7 (extreme value theorem). Let $f : X \rightarrow Y$ be continuous, where Y is an ordered set in the order topology. If X is compact, then there exist points c and d in X such that $f(c) \leq f(x) \leq f(d)$ for every $x \in X$.

Proof. Since f is continuous and X is compact, the set $A = f(X)$ is compact. We show that A has a largest element M and a smallest element m . Then since m and M belong to A , we must have $m = f(c)$ and $M = f(d)$ for some points c and d of X . If A has no largest element, then the collection $\{(-\infty, a) | a \in A\}$ forms an open covering of A . Since A is compact, some finite subcollection $\{(-\infty, a_1), \dots, (-\infty, a_n)\}$ covers A . If a_i is the largest of the elements $a_1, \dots, a_n$, then a_i belongs to none of these sets, contrary to the fact that they cover A . A similar argument shows that A has a smallest element. $\square$

Vocabulary 5.7 (Lebesgue number). Let X be a metric space, a Lebesgue number of X with respect to an open cover $\mathcal{A}$ of X is a number $\delta > 0$ such that for all $S \subseteq X$ of diameter $< \delta$ is contained in at least an element of $\mathcal{A}$.

Theorem 8 (Lebesgue Theorem). If (X, d) is a compact metric space, then for any open cover $\mathcal{A}$ of X , there exists a Lebesgue number $\delta > 0$.

Proof. Let $\mathcal{A}$ be an open covering of X . If X itself is an element of $\mathcal{A}$, then any positive number is a Lebesgue number for $\mathcal{A}$. So assume X is not an element of $\mathcal{A}$. Choose a finite subcollection $\{A_1, \dots, A_n\} \subseteq \mathcal{A}$ that covers X . For each i , set $C_i = X \setminus A_i$, and define $f : X \rightarrow \mathbb{R}$ by letting $f(x)$ be the average of the numbers $d(x, C_i)$

$$f(x) = \frac{1}{n} \sum_{i=1}^n d(x, C_i)$$

We show that $f(x) > 0$ for all x . Given $x \in X$, choose i so that $x \in A_i$. Then choose ϵ so the ϵ -neighborhood of x lies in A_i . Then $d(x, C_i) \geq \epsilon \rightarrow f(x) \geq \epsilon/n$. f is continuous implies the existence of minimum value δ ; we show that δ is the Lebesgue number. Let B be a subset of X of diameter less than δ . Choose a point x_0 of B ; then B lies in the δ -neighborhood of x_0 . Now

$$\delta \leq f(x_0) \leq d(x_0, C_m)$$

where $d(x_0, C_m)$ is the largest of the numbers $d(x_0, C_i)$. Then the δ -neighborhood of x_0 is contained in the element $A_m = X \setminus C_m$ of the covering $\mathcal{A}$. $\square$

Recall the definition of continuous: $f : X \rightarrow Y$, (X, d_x) , (Y, d_y) are metric spaces. f is continuous if and only if for all $x_0 \in X$ and $\epsilon > 0$, there exists $\delta > 0$ such that for all $x \in X$, $(d_X(x_0, x_1) < \delta$. This implies $d_Y(f(x_0), f(x_1)) < \epsilon$). Remember that delta depends on x_0 and ϵ .

Vocabulary 5.8 (uniformly continuous). f is called uniformly continuous if for every real number $\epsilon > 0$ there exists a real number $\delta > 0$ such that for every $x, y \in X$ with $d_1(x, y) < \delta$, we have $d_2(f(x), f(y)) < \epsilon$. The set $\{y \in X : d_1(x, y) < \delta\}$ for each x is a neighborhood of x and the set $\{x \in X : d_1(x, y) < \delta\}$ for each y is a neighborhood of y by the definition of a neighborhood in a metric space.

Theorem 9. Let $f : X \rightarrow Y$ be a continuous map of the compact metric space (X, d_X) to the metric space (Y, d_Y) . Then f is uniformly continuous.

Proof. Given $\epsilon > 0$, take the open covering of Y by balls $B(y, \epsilon/2)$ of radius $\epsilon/2$. Let $\mathcal{A}$ be the open covering of X by the inverse images of these balls under f . Choose δ to be a Lebesgue number for the covering $\mathcal{A}$. Then if x_1 and x_2 are two points of X such that $d_X(x_1, x_2) < \delta$, the two-point set $\{x_1, x_2\}$ has diameter less than δ , so that its image $\{f(x_1), f(x_2)\}$ lies in some ball $B(y, \epsilon/2)$. Then $d_Y(f(x_1), f(x_2)) < \epsilon$, as desired. $\square$

Vocabulary 5.9 (sequential compact). For any sequence (x_i) in X , there exists a subsequence (x_{i_j}) that converges to some point $x \in X$.

Vocabulary 5.10 (limit point compactness). Every infinite subset of X has a limit point.

Example. $\mathbb{R}^2$ is not sequentially compact. Take $x_i = (i, 0)$ is not bounded.

Theorem 10. Let X be a metrizable topological space, then X is compact if and only if X is sequentially compact.

Proof. This is a combination of 28.1 and 28.2 of the textbook.

($\Rightarrow$) **First Direction**

Step 1: Show that compact implies limit point compact.

Suppose A has no limit point, $S = \{\text{set of limit point}\}$, if $S = \emptyset$, $\emptyset \subseteq A \rightarrow A$ contains all of its limit points. This means A is closed. For each $a \in A$, can choose an open neighborhood U_a such that $U_a \cap A = \{a\}$ because A has no limit points. X is covered by open sets $X \setminus A$ and U_a . Since X is compact, there exists a finite subcover to this "combined" cover. Notice that $(X \setminus A) \cap A = \emptyset \rightarrow$ each U_a only contain one point of $A \rightarrow A$ is finite.

Step 2: Show limit point compact implies sequential compact.

Assume that X is limit point compact. Given a sequence (x_n) of points of X , consider the set $A = \{x_n | n \in \mathbb{Z}_+\}$. If the set A is finite, then there is a point x such that $x = x_n$ for infinitely many values of n . Thus there exists a converging (constant) subsequence of (x_n) . If A is infinite, then A has a limit point x by step 1. We obtain a subsequence of (x_n) converging to x by shrinking the ϵ -ball around x .

($\Leftarrow$) **Second Direction**

Step 1: Show that if X is sequentially compact, then the Lebesgue number lemma holds for X .

Let $\mathcal{A} = \{S | S \text{ open in } X\}$ be an open cover of X , suppose no $\delta > 0$ such that each set of diameter $< \delta$ is contained in an element of $\mathcal{A}$. Taking $\delta = 1/n$ a positive integer, there exists $C_n \subseteq X$ with diameter $1/n < \delta$ is not contained in any element of $\mathcal{A}$. Choose a point $x_n \in C_n$, for each n . By hypothesis, X is sequentially compact: some subsequence (x_{n_i}) of the sequence $(x_n) \rightarrow a$, for some $a \in X$. Then $a \in S \in \mathcal{A}$. Since S is open, can choose $\epsilon > 0$ such that $B(a, \epsilon) \subset S$. Now we need to carefully choose the value for i

(a) i is large enough so that C_{n_i} is in the $\epsilon/2$ neighborhood of x_{n_i} : makes sure that C_{n_i} is no more than $\epsilon/2$ away from the subsequence.

(b) i is large enough such that $d(x_{n_i}, a) < \epsilon/2$: makes sure that the sequence is no more than $\epsilon/2$ away from a .

But this means that $C_{n_i} \subset B(a, \epsilon) \subset S$, contrary to hypothesis.

Step 2: Show that if X is sequentially compact, then for any $\epsilon > 0$, there exists a finite covering of X by open ϵ -balls.

Suppose there exists some $\epsilon > 0$ such that X cannot be covered by finitely many ϵ -balls, then pick any point $x_1 \in X$, $B(x_1, \epsilon) \subsetneq X$ otherwise X could be covered by a single ϵ -ball. Choose $x_2 \notin B(x_1, \epsilon)$. Inductively, choose

$$x_{n+1} \notin B(x_1, \epsilon) \cup \dots \cup B(x_n, \epsilon)$$

since these balls do not cover X . Note that by construction $d(x_{n+1}, x_i) \geq \epsilon$ for $i = 1, \dots, n$. Therefore, the sequence (x_n) can have no convergent subsequence. In particular, X is not sequentially compact.

Step 3: Show that if X is sequentially compact, then X is compact.

Let $\mathcal{A}$ be an open covering of X . Because X is sequentially compact, the open covering $\mathcal{A}$ has a Lebesgue number δ by step 1. Let $\epsilon = \delta/3$. By step 2, we can use sequential compactness of X to find a finite covering of X by open ϵ -balls. Each of these balls has diameter at most $2 \cdot (\delta/3)$, so it lies in an element of $\mathcal{A}$. Choosing one such element of $\mathcal{A}$ for each of these finitely many ϵ -balls, we obtain a finite subcollection of $\mathcal{A}$ that covers X . $\square$

Lemma. Additionally, we can assume Y is Hausdorff and f is injective.

Tikhonov Theorem

It's Tikhonov babyyyy!

Remark. We have shown that the finite product of compact spaces is compact, now we check if arbitrary products work the same way.

Vocabulary 6.1 (finite intersection property (FIP)). A collection of subsets $\{C_a\}$ of a set X has the FIP if any finite intersection of this collection is not empty. More precisely, for any finite subcollection $\{C_{a_1}, \dots, C_{a_n}\} \subseteq \{C_a\}$, $C_{a_1} \cap \dots \cap C_{a_n} \neq \emptyset$.

Theorem 11 (Tikhonov). Any product of compact spaces is compact.

To prove this theorem, consider the following two lemmas.

Lemma. There exists a maximal collection of subsets with FIP.

Proof. Let X be a set. Let A and B be collections of subsets of X with FIP. Define $\mathcal{A} = \{B \mid A \subseteq B\}$. If $\mathcal{B} \subseteq \mathcal{A}$ is simply ordered by inclusion, let $C = \bigcup_{B \in \mathcal{B}} B$. For all $S \in \mathcal{B}$, $A \subseteq S$. Let $S_1, \dots, S_n$ be elements of C . For all i , $S_i \in B_i \in \mathcal{B}$ and $\{B_1, \dots, B_n\} \subseteq \mathcal{B}$, so it is simply ordered by the relation of proper inclusion. Being finite, it has a largest element, that is, there is an index k such that $B_i \subseteq B_k$ for $1 \leq i \leq n$. Then all sets $S_1, \dots, S_n$ are elements of B_k . Since B_k has the FIP, the intersection of such sets is empty. Therefore, C has the FIP. By Zorn's lemma, C is the maximal element of $\mathcal{A}$. $\square$

Lemma. Let X be a set and let M be a collection of subsets of X that is maximal with respect to the FIP, then

- (a) any finite intersection of elements of M is an element of M ;
- (b) if A is a subset of X that intersects every element of M , then A is an element of M .

Proof. (a) Let B be the intersection of finitely many elements of M . Define a collection $E = M \cup \{B\}$. Pick finitely many elements of E , if none of them is the set B , then their intersection is nonempty; if one of them is the set B , then their intersection is of the form $D_1 \cap \dots \cap D_m \cap B$, where $D_i \in M$, which is empty. (b) Given A , define $E = M \cup \{A\}$. Pick finitely many elements of E , if none of them is the set A , their intersection is nonempty; otherwise, it is of the form $D_1 \cap \dots \cap D_n \cap A$. By (a), $D_1 \cap \dots \cap D_n \in M$, so the intersection is nonempty. $\square$

Now we prove Tikhonov's theorem.

proof of the theorem. Let a collection of X_a be compact, define $X = \prod_{a \in J} X_a$. Let $\mathcal{A}$ be a collection of subsets of X with the FIP. By the first lemma, choose a collection $\mathcal{D}$ of subsets of X such that $\mathcal{A} \subseteq \mathcal{D}$ and $\mathcal{D}$ is maximal with the FIP. Given $a \in J$, let $\pi_a : X \rightarrow X_a$ be the projection map. Since $\mathcal{D}$ has the FIP, so does $\{\pi_a(D) \mid D \in \mathcal{D}\}$. For each a , choose a point $x_a \in X_a$ such that $x_a \in \bigcup_{D \in \mathcal{D}} \pi_a(D)$. Let $x = (x_a)_{a \in J} \in \prod_{a \in J} X_a$. The set U_b is a neighborhood of x_b in X_b . Since $x_b \in \overline{\pi_b(D)}$, there exists a point $\pi_b(y)$, where $y \in D$, such that $\pi_b(y) \in U_b \cap \pi_b(D)$. Then it follows that $y \in \pi^{-1}(U_b) \cap D$. By the second lemma, every subbasis element containing x belongs to $\mathcal{D}$, so every basis element containing x belongs to $\mathcal{D}$. Since $\mathcal{D}$ has the FIP, every basis element containing x intersects every element of $\mathcal{D}$, hence $x \in \overline{D}$ for every $D \in \mathcal{D}$ as desired. $\square$

There are many ways to prove Tikhonov's theorem, check the appendix for the proof by ultrafilters.

Compactification

Defined compactifications and what it means to have equivalent compactifications. Stone-Ćech Compactification. Discussed the uniqueness of compactifications under certain conditions and addressed how continuous functions extend over compactifications.

Vocabulary 7.1 (compactification). A compactification of a space X is a compact Hausdorff space Y containing X as a subspace such that $\overline{X} = Y$. Two compactifications Y_1 and Y_2 of X are said to be equivalent if there is a homeomorphism $h : Y_1 \rightarrow Y_2$ such that $h(x) = x$ for every $x \in X$.

Lemma. Let X be a space. Suppose that $h : X \rightarrow Z$ is an embedding of X in the compact Hausdorff space Z . Then there exists a corresponding compactification Y of X ; it has the property that there is an embedding $H : Y \rightarrow Z$ that equals h on X . Then the compactification Y is unique.

Proof. Given h , let $X_0 = h(X) \subseteq Z$, and let $Y_0 = \overline{X_0}$. Then Y_0 is a compact Hausdorff space and $\overline{X_0} = Y_0$, so Y_0 is a compactification of X_0 . Choose S such that $S \cap X = \emptyset$ and a map $k : S \rightarrow Y_0 \setminus X_0$ such that k is bijective. Let $Y = X \cup S$ and define $H : Y \rightarrow Y_0$ such that

$$H(y) = \begin{cases} h(y) & \text{for } y \in X \\ k(y) & \text{for } y \in S \end{cases}$$

We then give one more property to Y to give it a topology. Suppose that U is a subset of Y . Then U may only be open in Y if and only if $H(U)$ is open in Y_0 , hence H is a homeomorphism. It is also true that $H(y) = h(y)$ for all $y \in X$ and so we can extend the range of H to get an appropriate embedding $H : Y \rightarrow Z$. Suppose Y_i is a compactification of X such that $H_i : Y_i \rightarrow Z$ is an embedding that is an extension of h for $i \in \{1, 2\}$. Now H_i maps X onto $h(X) = X_0$. Because H_i is continuous, it maps Y_i into $\overline{X_0}$; since $H_i(Y_i)$ contains X_0 and is closed, it contains $\overline{X_0}$. Hence $H_i(Y_i) = \overline{X_0}$, and $H^{-1} \circ H_1$ defines a homeomorphism of Y_1 with Y_2 that equals the identity on X . $\square$

Vocabulary 7.2 (Stone-Čech compactification). The Stone-Čech compactification of X is a compact Hausdorff space $\beta(X)$ containing X as a subspace, such that $\overline{X} = \beta(X)$ and for any bounded continuous function $f : X \rightarrow \mathbb{R}$, there exists a bounded continuous extension $f' : \beta(X) \rightarrow \mathbb{R}$.

Vocabulary 7.3 (completely regular). A space X is completely regular if each one-point subset of X is closed and for any $x \in X$ and any closed set $C \subseteq X$, if $x \notin C$, then exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(c) = 0$ for all $c \in C$ and $f(x) = 1$.

Theorem 12 (Existence of Stone-Čech compactification). Let X be a completely regular space, there exists a compactification Y of X with the property that every bounded continuous map $f : X \rightarrow \mathbb{R}$ extends uniquely to a continuous map Y into $\mathbb{R}$.

Proof. Let $\{f_a\}_{a \in J}$ be the set of all bounded real functions. Define the real intervals $I_a = [\inf(f_a(X)), \sup(f_a(X))]$. These intervals are compact, thus by Tikhnov $\prod_{a \in J} I_a$ is also compact. Define $h : X \rightarrow \prod_{a \in J} I_a$ such that $h(x) = (f_a(x))_{a \in J}$. By the previous lemma, there exists H of the compactification of X induced by h that satisfies the previous lemma such that $H : Y \rightarrow \prod I_a$. Let f be any bounded continuous real function, then $f \in \{f_a\}_{a \in J}$. Since $f \in \{f_a\}_{a \in J}$, there exists b such that $f = f_b$. Define $\varphi_b : \prod I_a \rightarrow I_b$, then $\varphi_b \circ H : Y \rightarrow \mathbb{R}^J \rightarrow \mathbb{R}$, which is continuous. Since $H|_X = h$, we have $\varphi_b(H(x)) = \varphi_b(h(x)) = \varphi_b(f_a(x))_{a \in J} = f_b(x)$. Thus, we are able to construct an appropriate extension for any real-valued function that is bounded and continuous over X . $\square$

Theorem 13 (Uniqueness of Stone-Čech compactification). Let $A \subseteq X$; let $f : A \rightarrow Z$ be a continuous map of A into the Hausdorff space Z . There is at most one extension of f to a continuous function $g : \overline{A} \rightarrow Z$.

Proof. Let g and g' be two extensions of f . Pick x such that $g(x) \neq g'(x)$. Let U and U' be disjoint neighborhoods of $g(x)$ and $g'(x)$, respectively. Choose a neighborhood V of x so that $g(V) \subseteq U$ and $g'(V) \subseteq U'$, V intersects A at some point y , then $g(y) \in U$ and $g'(y) \in U'$. Since $y \in A$, $g(y) = f(y)$ and $g'(y) = f(y)$. This contradicts the fact that U and U' are disjoint. $\square$

Complete Metric Space

Defined Cauchy sequences. Showed that every compact metric space is complete. Uniform metric: metrics for function spaces. Compact convergence topology. Went over function space topologies and proved Arzelà-Ascoli.

Vocabulary 8.1 (Cauchy sequence). A sequence (x_n) of real numbers is called Cauchy, if for every positive real number ϵ , there is a positive integer $N \in \mathbb{N}$ such that for all natural numbers $m, n > N$, $|x_m - x_n| < \epsilon$.

Vocabulary 8.2 (complete). A metric space is complete if any Cauchy sequence in X converges in X .

Lemma. Let (x_i) be a Cauchy sequence in (X, d) . Then the followings are equivalent

1. (x_i) converges in X .
2. There is a converging subsequence of (x_i) in X .

Proof. (1) $\Rightarrow$ (2): take the subsequence to be itself. (2) $\Rightarrow$ (1): Let (x_{n_i}) be a subsequence converging to some $x \in X$. Take any $\epsilon > 0$, since (x_n) is Cauchy, there exists some $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_m, x_n) < \epsilon/2$. Since $x_{n_i} \rightarrow x, i \rightarrow \infty$, there exists an index $i \in \mathbb{N}$ such that $n_i > N$ and $d(x_{n_i}, x) < \epsilon/2$. Take $m = n_i$, we then have $d(x_n, x) \leq d(x_n, x_{n_i}) + d(x_{n_i}, x) < \epsilon/2 + \epsilon/2 = \epsilon$ concludes that $x_n \rightarrow x$. $\square$

Theorem 14. Every compact metric space is complete.

Proof. Note that in metric spaces the notions of compactness and sequential compactness coincide. Let x_n be a Cauchy sequence in the metric space X . Since X is sequentially compact there is a convergent subsequence $x_{n_k} \rightarrow x \in X$. It remains to show that $x_n \rightarrow x$. Since $x_{n_k} \rightarrow x$ there is N_1 with $n_k \geq N_1$ implies $d(x_{n_k}, x) < \varepsilon/2$. Let N_2 be such that $n, m \geq N_2$ implies $d(x_n, x_m) < \varepsilon/2$. Let $n \geq N = \max(N_1, N_2)$ and pick some $n_k \geq N$. Then

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon.$$

Hence X is complete. $\square$

Lemma. Every Cauchy sequence is bounded.

Proof. Let (a_n) be Cauchy. We choose $0 < \epsilon_0$. So for all $n > m \geq N_0$ we have that $|a_n - a_m| < \epsilon_0$. Therefore (a_n) is bounded for all $m \geq N_0$ by ϵ_0 . Since $\mathbb{N}_{N_0}$ is finite, it is bounded. So, for all $m < N_0$, (a_n) is bounded. Therefore (a_n) is bounded. $\square$

Theorem 15. $\mathbb{R}^n$ is complete for each $n \in \mathbb{N}$ (with respect to any metric d_p for any $p \in [1, \infty]$).

Proof. Take any Cauchy sequence (x_n) in $\mathbb{R}^n$. It is bounded. So $(x_n)_{n \in \mathbb{N}}$ is contained in some closed ball (with respect to d_p). This ball is closed and bounded in $\mathbb{R}^n$, thus compact. This implies sequentially compact. It follows that there exists a subsequence of (x_n) converging to some x in a ball. This implies (x_n) converges to $x \in \mathbb{R}^n$. $\square$

Vocabulary 8.3 (bounded metric). $\bar{d}(x, y) = \min\{d(x, y), 1\}$ is the bounded metric.

Vocabulary 8.4 (uniform metric). Set $\vec{x} = (x_\alpha)_{\alpha \in J}$ and $\vec{y} = (y_\alpha)_{\alpha \in J}$. $\vec{x}$ and $\vec{y}$ are points of the Cartesian product Y^J . Define the uniform metric as $\bar{\rho}(\vec{x}, \vec{y}) = \sup\{\bar{d}(x_\alpha, y_\alpha) | \alpha \in J\}$. When viewing Y^J as functions from J to Y , we can define uniform metric corresponding to the metric $\bar{d}$ on Y as $f, g : J \rightarrow Y$, $\bar{\rho} = \sup\{\bar{d}(f(\alpha), g(\alpha)) | \alpha \in J\}$. We can check $\bar{\rho}$ is indeed a metric.

Vocabulary 8.5 (open with respect to the uniform metric). For any $x \in X$, $U \subseteq Y$ is open if $S(x, U) := \{f : X \rightarrow Y | f(x) = U\} \subseteq Y^X$.

Theorem 16. If the space Y is complete in the metric d , then the space Y^J is complete in the uniform metric $\bar{\rho}$ corresponding to d .

Proof. Observe that $\bar{d}(x, y) \leq d(x, y)$, then let (y_i) be a sequence in Y . By the inequality, (y_i) is Cauchy with respect to d implies (y_i) is Cauchy with respect to $\bar{d}$. Moreover, (y_i) converges in Y with respect to d . It follows that (y_i) converges in Y with respect to $\bar{d}$. It follows that if (Y, d) is complete then $(Y, \bar{d})$ is complete. Let (f_i) be a Cauchy sequence in Y^J relative to $\bar{\rho}$. If we think of (f_i) as functions, then the difference at a single point is less or equal to the supremum of all differences, hence we have the following inequality

$$\bar{d}(f_n(\alpha), f_m(\alpha)) \leq \bar{\rho}(f_n, f_m)$$

for all m, n . This shows $(f_i(\alpha))$ is Cauchy relative to $\bar{d}$. Then $(f_i(\alpha)) \rightarrow y_\alpha$ to some $y_\alpha \in Y$. Let $f \in Y^J$ such that $f(\alpha) = y_\alpha$. Given $\epsilon > 0$, first choose N large enough that $\bar{\rho}(f_n, f_m) < \epsilon/2$ whenever $n, m \geq N$. Then $\bar{d}(f_n(\alpha), f_m(\alpha)) < \epsilon/2$. Since $(f_i) \rightarrow f$ at α , if we let m large, $\bar{d}(f_n(\alpha), f(\alpha)) \leq \epsilon/2$. Let f be the function that does this for all α when $n \geq N$, we have

$$\bar{\rho}(f_n, f) \leq \epsilon/2 < \epsilon$$

This proves that $(f_n) \rightarrow f$ as required. $\square$

Vocabulary 8.6 (set of continuous/bounded functions). A function f is said to be bounded if its image $f(X)$ is a bounded subset of the metric space (Y, d) . $C(X, Y)$ and $B(X, Y)$ are the set of continuous and the set of bounded functions respectively.

Theorem 17. Let X be a topological space and let (Y, d) be a metric space. The set $C(X, Y)$ of continuous functions is closed in Y^X under the uniform metric. So is the set $B(X, Y)$ of bounded functions. Therefore, if Y is complete, these spaces are complete in the uniform metric.

Proof. We recall the uniform limit theorem to prove a part of the theorem.

Part 1: if the sequence $(f_n) \subset Y^X$ converges to some element $f \in Y^X$ relative to $\bar{\rho}$, then for any $\epsilon > 0$, choose some N such that $\bar{\rho}(f_n, f) < \epsilon$, for all $n > N$. Since $\bar{d}$ is bounded by $\bar{\rho}$, we have

$$\bar{d}(f_n(x), f(x)) \leq \bar{\rho}(f_n, f) < \epsilon \Rightarrow (f_n) \rightarrow f$$

Part 2: let $f \in Y^X$ be any limit point of $C(X, Y)$. Then there is a sequence $(f_n) \subset C(X, Y)$ such that $(f_n) \rightarrow f$ relative to $\bar{\rho}$. By the uniform limit theorem, the limit should also be a continuous function, thus $f \in C(X, Y)$. Then $C(X, Y)$ contains all limit points. It is closed.

Part 3: let $f \in Y^X$ be any limit point of $B(X, Y)$. Then there is a sequence $(f_n) \subset B(X, Y)$ such that $(f_n) \rightarrow f$. Since convergent, there exists some N such that $\bar{\rho}(f_N, f) < 1/2$. Then for $x \in X$, we have $\bar{d}(f_N(x), f(x)) < 1/2$, which implies that $d(f_N(x), f(x)) < 1/2$. It follows that if M is the diameter of the set $f_N(X)$, then $f(X)$ has diameter at most $M + 1$. Hence $f \in B(X, Y)$. We conclude that $C(X, Y)$ and $B(X, Y)$ are complete in the metric $\bar{\rho}$ if Y is complete in d . $\square$

Vocabulary 8.7 (sup metric). If (Y, d) is a metric space, one can define another metric on the set $B(X, Y)$ by the equation $\rho(f, g) = \sup\{d(f(x), g(x)) | x \in X\}$. It is easy to see that ρ is well-defined(?), for the set $f(X) \cup g(X)$ is bounded if both $f(X)$ and $g(X)$ are. The metric ρ is called the sup metric.

Remark. For $f, g \in B(X, Y)$, we have $\bar{\rho}(f, g) = \min\{\rho(f, g), 1\}$.

Theorem 18. Let (X, d) be a metric space. There is an isometric embedding of X into a complete metric space.

Proof. Let $B(X, \mathbb{R})$ be the usual stuff. Let x_0 be a fixed point of X . Define

$$\phi_a : X \rightarrow \mathbb{R} \quad \phi_a(x) = d(x, a) - d(x, x_0)$$

From triangle inequality we have $d(x, a) \leq d(x, b) + d(a, b)$ and $d(x, b) \leq d(x, a) + d(a, b)$. By subtraction, $|d(x, a) - d(x, b)| \leq d(a, b)$. Setting $b = x_0$ we showed $|\phi_a(x)| \leq |d(a, x_0)|$ for all x . $\phi_a(x)$ is bounded. Define

$$\Phi : X \rightarrow B(X, \mathbb{R}) \quad \Phi(a) = \phi_a$$

We show that Φ is an isometric embedding of (X, d) into the complete metric space $(B(X, \mathbb{R}), \rho)$. By definition,

$$\rho(\phi_a, \phi_b) = \sup\{|\phi_a - \phi_b|\} = \sup\{|d(x, a) - d(x, b)|\} \leq \sup\{d(a, b)\} = d(a, b)$$

Notice that the inequality $|d(x, a) - d(x, b)| \leq d(a, b)$ cannot be strict when $x = a$. Thus we have

$$\rho(\phi_a, \phi_b) = \sup\{|d(x, a) - d(x, b)|\} = \sup\{d(a, b)\} = d(a, b)$$

which shows $\rho(\phi_a, \phi_b) = d(a, b)$ proving Φ is an isometric embedding. $\square$

Vocabulary 8.8 (completion). Let X be a metric space. If $h : X \rightarrow Y$ is an isometric embedding of X into a complete metric space Y , then the subspace $h(X)$ of Y is a complete metric space. It is called the completion of X .

Vocabulary 8.9 (pointwise convergence). Given a point $x \in X$ and an open set $U \subset Y$, let $S(x, U) = \{f | f \in Y^X \text{ and } f(x) \in U\}$. The sets $S(x, U)$ are a subbasis for topology on Y^X , which is called the topology of pointwise convergence.

Remark. The general basis element for this topology is a finite intersection of subbasis elements $S(x, U)$. Thus a typical basis element about the function f consists of all functions g that are "close" to f at finitely many points.

Vocabulary 8.10 (compact convergence topology). Let (Y, d) be a metric space. Let X be a topological space. Given an element $f \in Y^X$, a compact subspace $C \subset X$, and a number $\epsilon > 0$, let $B_C(f, \epsilon)$ denote the set of all those elements $g \in Y^X$ for which

$$\sup\{d(f(x), g(x)) | x \in C\} < \epsilon$$

The sets $B_C(f, \epsilon)$ form a basis for a topology on Y^X . It is called the topology of compact convergence.

Lemma. Check that $\mathcal{B}_{cpt} := \{B_C(f, \epsilon)\}$ form a basis of Y^X .

Proof. Check both criteria ① for any $h \in Y^X$, let C be some compact set that h passes through. Pick an appropriate $\epsilon > 0$ we have $h \in B_C(h, \epsilon)$. ② let $f \in B_{C_1}(f_1, \epsilon_1) \cap B_{C_2}(f_2, \epsilon_2)$. We choose $C_3 = C_1 \cap C_2$, $f_3 = f$, and $\epsilon_3 = \min\{\epsilon_1, \epsilon_2\}$. We conclude that $\mathcal{B}_{cpt}$ is a basis and $\mathcal{T}_{cpt}$ is the topology generated by $\mathcal{B}_{cpt}$. $\square$

Vocabulary 8.11 (uniform convergence topology). Define

$$B_X(f, \epsilon) = \{g : X \rightarrow Y \mid \sup_{x \in X} \{d(f(x), g(x))\} < \epsilon\}$$

for $f \in Y^X, \epsilon > 0$ as basis elements. We get the basis for the uniform convergence topology $\mathcal{B}_{uni} = \{B_X(f, \epsilon)\}$. Comparing to basis elements of $\mathcal{B}_{cpt}$

$$B_C(f, \epsilon) = \{g : X \rightarrow Y \mid \sup_{x \in C} \{d(f(x), g(x))\} < \epsilon\}$$

Instead of requiring it to stay close on all of X , B_X only requires it to stay close in a compact subset. Thus $\mathcal{T}_{cpt} \subset \mathcal{T}_{uni}$. Meanwhile, $\bigcap_i S(x_i, U_i)$ only requires it to stay close near finitely many points hence $\mathcal{T}_{pt} \subset \mathcal{T}_{cpt}$.

Theorem 19. Let X be a space, and let (Y, d) be a metric space. For the function space Y^X , the following inclusions hold:

$$\mathcal{T}_{pt} \subset \mathcal{T}_{cpt} \subset \mathcal{T}_{uni}$$

If X is compact, the last two coincide, and if X is discrete, the first two coincide.

Proof. Clear. $\square$

Notation. Bases of uniform convergence topology, compact convergence topology, and pointwise convergence topology are denoted as $\mathcal{B}_{uni}$, $\mathcal{B}_{cpt}$, and $\mathcal{B}_{pt}$. Topologies generated by these bases are denoted as $\mathcal{T}_{uni}$, $\mathcal{T}_{cpt}$, and $\mathcal{T}_{pt}$.

Vocabulary 8.12 (compact-open topology). Let X and Y be topological spaces. If C is a compact subspace of X and U is an open subset of Y , define

$$S(C, U) = \{f \mid f \in C(X, Y) \text{ and } f(C) \subset U\}.$$

The sets $S(C, U)$ form a subbasis for a topology on $C(X, Y)$ that is called the compact-open topology. Comparing to the pointwise convergence topology

$$S(x, U) = \{f \mid f \in Y^X \text{ and } f(x) \in U\}$$

we see that the compact-open topology is finer.

Theorem 20. Let X be a space and let (Y, d) be a metric space. On the set $C(X, Y)$, the compact-open topology and the topology of compact convergence coincide.

Lemma. If K is compact subset of Y and V is an open subset of Y such that $K \subset V$. Then there exist an $\epsilon > 0$ such that $U(K, \epsilon) \subset V$.

proof of the lemma. Consider the function $f : K \rightarrow \mathbb{R}$ defined by $f(x) = d(a, Y - V)$. Since the domain K is compact, by the extreme value theorem, there is some point $x \in K$ at which the function f attains its minimum value ϵ . In a metric space, for every point x and subset A , it can be easily proven that

$$d(x, A) = 0 \iff x \in \bar{A}.$$

Note that V is an open set, so its complement $Y - V$ is a closed set and hence $\overline{Y - V} = Y - V$. As we can see, we have $K \subseteq V$; that is, $K \cap (Y - V) = \emptyset$, so $f(x) > 0$ for all $x \in K$. That is why $\epsilon > 0$. Then we need to show that $U(K, \epsilon) \subseteq V$.

We can prove it by contradiction. Suppose $y \in U(K, \epsilon)$ but $y \notin V$. Then for every $x \in K$, we have

$$d(x, y) \geq d(x, Y - V) \geq \epsilon \implies d(y, K) = \inf_{x \in K} d(y, x) \geq \epsilon$$

producing a contradiction. $\square$

Proof of the theorem. If A is a subset of Y and $\epsilon > 0$, let $U(A, \epsilon)$ be the ϵ -neighborhood of A . If A is compact and V is an open set containing A , then there is an $\epsilon > 0$ such that $U(A, \epsilon) \subset V$. Indeed, the minimum value of the function $d(a, X \setminus V)$ is the required ϵ . (This is proven as the above lemma)

We first prove that the topology of compact convergence is finer than the compact-open topology. Let $S(C, U)$ be a subbasis element for the compact-open topology, and let $f \in S(C, U)$. Because f is continuous, $f(C)$ is a compact subset of the open set U . Therefore by the above, we can choose ϵ so that ϵ -neighborhood of $f(C) \subset U$. Then $B_C(f, \epsilon) \subset S(C, U)$ as desired.

Now we prove that the compact-open topology is finer than the topology of compact convergence. Let $f \in C(X, Y)$. Given an open set about f in the topology of compact convergence, it contains a basis element of the form $B_C(f, \epsilon)$. We shall find a basis element for the compact-open topology that contains f and lies in $B_C(f, \epsilon)$.

Each point $x \in X$ has a neighborhood V_x such that $f(\overline{V_x})$ lies in an open set U_x of Y having diameter less than ϵ . [For example, choose V_x so that $f(V_x)$ lies in the $\epsilon/4$ -neighborhood of $f(x)$. Then $f(\overline{V_x})$ lies in the $\epsilon/3$ -neighborhood of $f(x)$, which has diameter at most $2\epsilon/3$.] Cover C by finitely many such sets V_x , say for $x = x_1, \dots, x_n$. Let $C = \overline{V_x} \cap C$. Then C is compact, and the basis element

$$S(C_{x_1}, U_{x_1}) \cap \dots \cap S(C_{x_n}, U_{x_n})$$

contains f and lies in $B_C(f, \epsilon)$, as desired. $\square$

Vocabulary 8.13 (relatively compact). Let X be a topological space. A subset $C \subseteq X$ is relatively compact if the closure $\overline{A}$ is compact.

Vocabulary 8.14 (equicontinuous). Let X be a top space, (Y, d) be a metric space $C(X, Y)$. Let $F \subseteq C(X, Y)$ be a family of continuous functions $X \rightarrow Y$. F is equicontinuous if for all $x_0 \in X$ and for all $\epsilon > 0$, there exists a neighborhood U of x_0 (in X) [such that for all $f \in F$, $f(U) \subseteq B(f(x_0), \epsilon)$] or [such that for all $f \in F$, for all $x \in U$ $d(f(x_0), f(x)) < \epsilon$].

Example. Let X be a topological space and (Y, d) be a metric space. Let E be the set of homomorphisms from X to Y . If A is an equicontinuous family from X to Y , then the subspace topology on A of E with compact-open topology is the same as the subspace topology on A of E with the product topology.

Lemma. Consider maps $j : C(X, Y) \rightarrow X \times C(X, Y)$ defined by $f \mapsto a \times f$ and $e : X \times C(X, Y) \rightarrow Y$ defined by $x \times f \mapsto f(x)$. j, e are continuous.

Proof. Clearly j is continuous. Given $(x, f) \in X \times C(X, Y)$ and an open set V in Y about the image point $e(x, f) = f(x)$. Since f is continuous and X is locally compact Hausdorff space, pick any open set U about x with compact closure $\overline{U}$, such that f carries $\overline{U}$ into V . Consider the open set $U \times S(\overline{U}, V)$ in $X \times C(X, Y)$, which is open and containing (x, f) , then $e(x', f')$ belongs to this set, so $e(x', f') = f'(x')$ belongs to V . $\square$

Theorem 21 (Arzela-Ascoli). Let X be a topological space, (Y, d) be a metric space and $F \subseteq C(X, Y)$. Put the compact convergence topology on $C(X, Y)$. Let F be a subset of $C(X, Y)$. If F is equicontinuous under d and the set $F_a = \{f(a) \mid f \in F\}$ has compact closure for each $a \in X$, then F is contained in a compact subspace of $C(X, Y)$. The converse holds if X is locally compact Hausdorff.

Proof. ($\Rightarrow$) This proof is adapted from Lang's Real and Functional Analysis. Let F be equicontinuous and for each $x \in X$, the set $F(x)$ consisting of all values $f(x)$ for $f \in F$ relatively compact. Let $r > 0$. By equicontinuity, for each $x \in X$, select an open neighborhood $V(x)$ such that, if $y \in V(x)$, then $|f(y) - f(x)| < r$ for all $f \in F$. Then we have a finite number of neighborhoods that covers X , namely, $V(x_1), \dots, V(x_n)$. Each set $F(x_1), \dots, F(x_n)$ is relatively compact, so $Y = \bigcup F(x_i)$ is compact. Let $B(a_1), \dots, B(a_m)$ be open balls of radius r centered at points $a_1, \dots, a_m$ covers Y , so $f(x_i) \in B(a_{\sigma_i})$ for $i \in \{1, \dots, n\}$ and $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, m\}$ is a map. For all σ , let F_σ be the set of $f \in F$ such that, for all i , $|f(x_i) - a_{\sigma_i}| < r$. Then the finite number of F_σ cover F . If $f, g \in F_a$ and $x \in X$, then there exists some neighborhood such that $x \in V(x_i)$, so $|f(x) - g(x)| \leq |f(x) - f(x_i)| + |f(x_i) - a_{\sigma_i}| + |a_{\sigma_i} - g(x_i)| + |g(x_i) - g(x)| < 4r$. Then F is covered by a finite number of balls of prescribed radius. ($\Leftarrow$) Let H be a compact subspace of $C(X, Y)$ contains F . By the lemma above, $e \circ j : H \rightarrow H_a$ is continuous. Since H is compact, H_a is compact as well. Let A be a compact subspace of X containing a neighborhood of a . Let $C(A, Y)$ be the compact convergence topology. Let f be an element of $C(X, Y)$ and let $B = B_C(f|_A, \epsilon)$ be a basis element of $C(A, Y)$ containing the restriction of f . So C is compact subspace of X , and the map $r : C(X, Y) \rightarrow C(A, Y)$ maps $B_C(f, \epsilon)$ of f in $C(X, Y)$ into B . Now we have $r : H \mapsto \{f|_A; f \in H\}$. Let this set be R , then R is compact subspace of $C(A, Y)$. Recall that A is compact, the compact convergence and uniform topologies on $C(A, Y)$ coincide, so R is totally bounded in $C(A, Y)$. Hence R is equicontinuous with respect to d . $\square$

Introduction to algebraic topology

This is a quick introduction to algebraic topology taught on the last week of the course. Only basic facts like homotopy, contraction, fundamental group, etc.

Vocabulary 9.1 (deformation retraction). The retraction that retracts X to A is a set of maps such that $f_t : X \rightarrow X$ where $t \in I$ represents time. We then have $f_{t=0}(X) = X$ as the identity map and $f_{t=1}(X) = A$. $f_t|_A$ is always the identity map. Moreover, the set f_t should be continuous in the sense that the associated map

$$m : X \times I \rightarrow X \quad (x, t) \mapsto f_t(x)$$

is continuous.

Vocabulary 9.2 (mapping cylinder). the quotient space of the disjoint union $(X \times I) \amalg Y$ obtained by identifying each $(x, 1) \in X \times I$ with $f(x) \in Y$. In other words, this map records every phase of the transformation.

Remark. If a space X deformation retracts onto a subspace A via $f_t : X \rightarrow X$, then if $r : X \rightarrow A$ denotes the resulting retraction and $i : A \rightarrow X$ the inclusion, we have $r \circ i = \text{id}$ and $i \circ r \simeq \text{id}$, the latter homotopy being given by f_t (the homotopy equivalence of i is f_t).

Vocabulary 9.3 (homotopy equivalence). A map $f : X \rightarrow Y$ is called a homotopy equivalence if there is a map $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}$ and $g \circ f \simeq \text{id}$. The spaces X and Y are said to be homotopy equivalent or to have the same homotopy type.

Vocabulary 9.4 (contractible). A space having the homotopy type of a point.

Vocabulary 9.5 (nullhomotopic). A map that is homotopic to a constant map.

Proposition 23. A space X is contractible if and only if its identity map is nullhomotopic.

Proof. Contractible implies it is homotopy equivalent to a point $\{p\}$. Then there exists $f : X \rightarrow \{p\}$ and $g : \{p\} \rightarrow X$ such that $f \circ g \simeq \text{id}_{\{p\}}$ and $g \circ f \simeq \text{id}_X$. Nullhomotopic means there exists a homotopy $F : X \times I \rightarrow X$ such that $F(x, 0) = \text{id}_X$ and $F(x, 1) = c$ for some constant c .

($\Rightarrow$) Suppose contractible, we can build the homotopy as such

$$F : X \times I \rightarrow \{\text{Deformation retraction from } X \text{ to } \{p\}\} \quad (x, t) \mapsto f_t(x)$$

where $f_0 = \text{id}_X$ and $f_1 = \text{id}_{\{p\}}$. We can do this because $f \circ g \simeq \text{id}_{\{p\}}$ and $g \circ f \simeq \text{id}_X$ allowed us to morph a $\{p\} \rightarrow \{p\}$ map into an $X \rightarrow X$ map and we can morph it back. Here the constant c is $\{p\}$.

($\Leftarrow$) Suppose you have a homotopy F and let $\{p\} \in X$ be the constant ($F(x, 1) = \{p\}, \forall x \in X$). Take $f : X \rightarrow \{p\}$ the only possible map and $g : \{p\} \rightarrow X$ the inclusion map. Then obviously $f \circ g$ is the identity of $\{p\}$. Now

$$g \circ f : X \rightarrow \{p\} \rightarrow X \quad x \mapsto \{p\}$$

so F is a homotopy between id_X and $g \circ f$ (it is given that we can morph id_X continuously into c); thus $g \circ f \simeq \text{id}_X$. $\square$

Remark. Note that homotopy equivalence does not require the composition $X \rightarrow \{p\} \rightarrow X$ to be the identity map. It only required the existence of a homotopy from id_X to a constant map (homotopic to id_X).

Vocabulary 9.6 (path homotopy). Let f and g be paths in a topological space X . A path homotopy from f to g is a homotopy that is stationary on $\{0, 1\} \subseteq I$. If there exists a path homotopy between f and g , then we say f and g are path-homotopic.

Let X be a topological space. For any points $p, q \in X$, path homotopy is an equivalence relation on the set of all paths in X from p to q . Let f be a path in X . The path homotopy equivalence class of f is called the path class, denoted $[f]$.

Vocabulary 9.7 (base point). Let f be a loop in X with an initial point p , then we say f is based on p and p is called the base point of f .

Vocabulary 9.8 (fundamental group). Let X be a topological space and let $p \in X$ be a point. The fundamental group of X based at p , denoted $\pi_1(X, p)$, is the set of all path classes of loops based at p .

Given two paths g and f , if g starts at the end of f , then we define the multiplication of paths $f \circ g$ to be the path obtained by first following f and then following g . It is not hard to check that $\pi_1(X, p)$ is a group.

Example. The fundamental group of the torus T^2 is $\mathbb{Z} \times \mathbb{Z}$.

Example. The fundamental group of $\mathbb{R}$ is $\{0\}$, the trivial group.

Example. The fundamental group of $S^1 \vee S^1$ is $\mathbb{Z} * \mathbb{Z}$.

Appendix

Appendix: Basic Category Theory

We have discussed some basic definitions of topological spaces. It is time to introduce another way to reformulate those basic definitions using universal properties.

Vocabulary 10.1 (category). A category is a system consists of:

1. a class of objects;
2. morphisms between objects;
3. operations $\text{dom}(f)$, $\text{cod}(f)$, id_A , $g \circ f$, where A is an object and f , g , and h are morphisms,

such that

1. $\text{dom}(\text{id}_A) = A$;
2. $\text{cod}(\text{id}_A) = A$;
3. $f \circ \text{id}_{\text{dom}(f)} = f$;
4. $\text{id}_{\text{cod}(f)} \circ f = f$;
5. $\text{dom}(g \circ f) = \text{dom}(f)$;
6. $\text{cod}(g \circ f) = \text{cod}(g)$;
7. $h \circ (g \circ f) = (h \circ g) \circ f$.

Example. In **Set**, objects are sets and morphisms are functions. In **Top**, objects are topological spaces and morphisms are continuous functions. For any category $\mathcal{C}$, the dual category $\mathcal{C}^{\text{op}}$ consists of all objects in $\mathcal{C}$ and the morphisms are reversed.

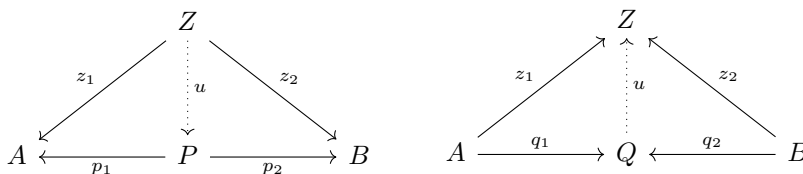
Vocabulary 10.2 (pullback and pushforward). Let $\mathcal{C}$ be a category and $X, Y, Z \in \text{Ob}(\mathcal{C})$. Let $f : X \rightarrow Y$ be a morphism. The pushforward of f is a set $f_* : \text{hom}(Z, X) \rightarrow \text{hom}(Z, Y)$ defined by $f_* : g \mapsto fg$. Similarly, the pullback of f is $f^* : \text{hom}(Y, Z) \rightarrow \text{hom}(X, Z)$ defined by $f^* : g \mapsto gf$.

Vocabulary 10.3 (isomorphism). Let $\mathcal{C}$ be a category and $f : A \rightarrow B$ a morphism in $\mathcal{C}$, then f is left invertible if there exists $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$ and g is a left inverse of f . Similarly, we can define right invertible and a right inverse. An isomorphism is a morphism which is both left and right invertible.

For all sentence Σ in the elementary language of category, the dual Σ^* is taken by changing:

$$\text{dom to cod} \quad \text{cod to dom} \quad f \circ g \text{ to } g \circ f$$

Vocabulary 10.4 (product). A diagram $A \xleftarrow{p_1} P \xrightarrow{p_2} B$ is a product if for all object Z and a diagram $A \xleftarrow{z_1} Z \xrightarrow{z_2} B$, there exists a unique $u : Z \rightarrow P$ such that $z_i = p_i \circ u_i$. The coproduct is the dual of product, that is, a diagram $A \xrightarrow{q_1} Q \xleftarrow{q_2} B$ is a coproduct if for all object Z and a diagram $A \xrightarrow{z_1} Z \xleftarrow{z_2} B$, there exists a unique $u : Q \rightarrow Z$ such that $z_i = q_i \circ u$.



Proposition 24. The disjoint union topology is a coproduct in **Top**.

Remark. For a composition of arrows $g \circ f$, we write gf for convenience.

Vocabulary 10.5 (functor). A functor F from a category C to D is a system consists of:

1. an object FX of the category D for each object X in the category C ;
2. a morphism $Ff : FX \rightarrow FY$ for every morphism $f : X \rightarrow Y$,

such that

1. $(Fg)(Ff) = F(gf)$ for any morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$;
2. $Fid_X = id_{FX}$ for any object X .

Vocabulary 10.6 (natural transformation). Let F and G be functors $C \rightarrow D$. A natural transformation η from F to G consists of a morphism $\eta_X : FX \rightarrow GX$ for each object X in C . Moreover, these morphisms in D must satisfy the property that $\eta_Y Ff = Gf \eta_X$ for every morphism $f : X \rightarrow Y$ in C .

We also need the notion of universal properties, which characterizes some object up to isomorphism. More intuitively, consider the Yoneda lemma.

Lemma (Yoneda). Let C be any category. For every object X in C and for every functor $F : C^{op} \rightarrow Set$, the set of natural transformations from $hom(-, X)$ to F is isomorphic to FX .

Given any category C , to actually say that an object in the category exists, you need a set and some explicit formula for how to perform the operations, and an universal property gives a unique presentation of it.

In the following paragraphs, we will give some better definitions and universal properties. The definitions are equivalent to previous definitions; and it is left to the readers to check.

Appendix: Several Universal Properties

Vocabulary 10.7 (subspace topology). Let $(X, \mathcal{T}_X)$ be a topological space, and $Y \subset X$. The subspace topology on Y is the coarsest topology on Y for which the canonical inclusion $i : Y \hookrightarrow X$ is continuous.

Universal property for the subspace topology. Let $(X, \mathcal{T}_X)$ be a topological space, let Y be a subset of X , and let $i : Y \hookrightarrow X$ be the natural inclusion. For every topological space $(Z, \mathcal{T}_Z)$ and every function $f : Z \rightarrow Y$, f is continuous if and only if $if : Z \rightarrow X$ is continuous.

$$\begin{array}{ccc} Z & \xrightarrow{\quad f \quad} & Y \\ & \searrow \scriptstyle if & \downarrow \scriptstyle i \\ & & X \end{array}$$

Vocabulary 10.8 (quotient topology). Let $(X, \mathcal{T}_X)$ be a topological space, let S be a set, and let $\pi : X \twoheadrightarrow S$ be surjective. The quotient topology on S is the finest topology for which π is continuous, and π is called a quotient map.

Universal property for the quotient topology. Let $(X, \mathcal{T}_X)$ be a topological space, let S be a set, and let $\pi : X \twoheadrightarrow S$ be surjective. For every topological space $(Z, \mathcal{T}_Z)$ and every function $f : S \rightarrow Z$, f is continuous if and only if $f\pi : X \rightarrow Z$ is continuous.

$$\begin{array}{ccc} X & \xrightarrow{\quad \pi \quad} & S \\ & \searrow \scriptstyle f\pi & \downarrow \scriptstyle f \\ & & Z \end{array}$$

Proposition 25. The dual of a subspace topology is the quotient topology.

We have shown the universal property of product and coproduct, we encourage readers to find the universal property of product and coproduct topologies.

Universal property for the Stone-Ćech compactification. For every map $f : X \rightarrow Y$ between X and a compact Hausdorff space Y , there is a unique map $U\hat{f} = \hat{f} : \beta X \rightarrow Y$, the adjunct of f , such that $\hat{f}\eta = f$.

$$\begin{array}{ccc} \beta X & \xleftarrow{\eta} & X \\ & \searrow \hat{f} & \downarrow f \\ & & Y \end{array}$$

Appendix: Ultrafilters

Vocabulary 10.9 (filter). Let X be a set. A filter on X is a family $\mathcal{F}$ of subsets such that:

1. $X \in \mathcal{F}$ and $\emptyset \notin \mathcal{F}$;
2. if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$;
3. if $A \subset B \subset X$ and $A \in \mathcal{F}$, then $B \in \mathcal{F}$.

Vocabulary 10.10 (ultrafilter). An ultrafilter on a set X is the maximal filter on X .

Vocabulary 10.11 (prime filter). A filter $\mathcal{P}$ on a set X is prime if it is proper and for all $A, B \subset X$, $A \cup B \in \mathcal{P}$ implies either $A \in \mathcal{P}$ or $B \in \mathcal{P}$.

Proposition 26. A filter is prime if and only if it is an ultrafilter.

Ultrafilter lemma. Every proper filter is contained in an ultrafilter.

Proposition 27. A space X is compact if and only if every prime filter converges.

Proposition 28. Let $\mathcal{U}$ be an ultrafilter on X and let $f : X \rightarrow Y$. The pushforward $f_*\mathcal{U}$ is an ultrafilter on Y .

Finally, let's prove Tikhonov's theorem by ultrafilters.

Proof. Let $\{X_\alpha\}_{\alpha \in A}$ be a family of compact spaces, define $X = \prod_{\alpha \in A} X_\alpha$ and let $\mathcal{F}$ be an ultrafilter on X . Since each X_α is compact, there exists x_α such that $(\pi_\alpha)_*\mathcal{F} \rightarrow x_\alpha$. For all open neighborhoods U of x_α , there exists $B \in \mathcal{F}$ with $\pi_\alpha B \subseteq U$. Equivalently, $B \subset \pi_\alpha^{-1}U$, so $\pi_\alpha^{-1}U \in \mathcal{F}$. Every open neighborhood of $(x_\alpha)_{\alpha \in A}$ is a union of finite intersections of some $\pi_\alpha^{-1}U$. Therefore, $\mathcal{F} \rightarrow (x_\alpha)_{\alpha \in A}$. $\square$

Appendix: Cell Complex

In algebraic topology, there exists a natural cellular homology on CW-complex which gives an isomorphism to the simplicial homology of a topological space. Essentially, this means that we can analyze the topological space via deconstruct it into some complex structures (simplexes and cells) and analyze the n -skeleton of each construction. Formally, this process is represented by two functors

$$\text{Top} \xrightarrow{C_\bullet} \text{Ch} \xrightarrow{H_*} \text{GrAb}.$$

This process allows one to turn an topological question into an algebraic one, which gives an natural description to describe certain topological invariants.

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